CSE 0612-2131

Trees and Binary Trees

Presented By
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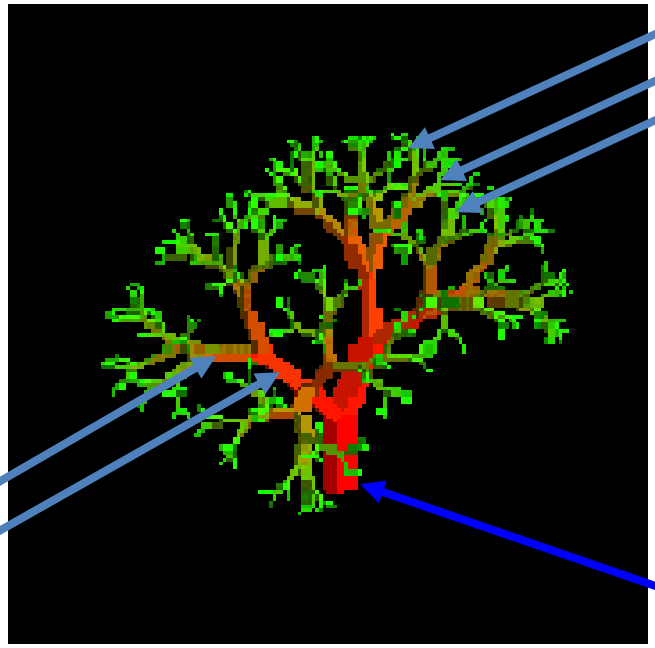
Dept. of CSE, NBIU

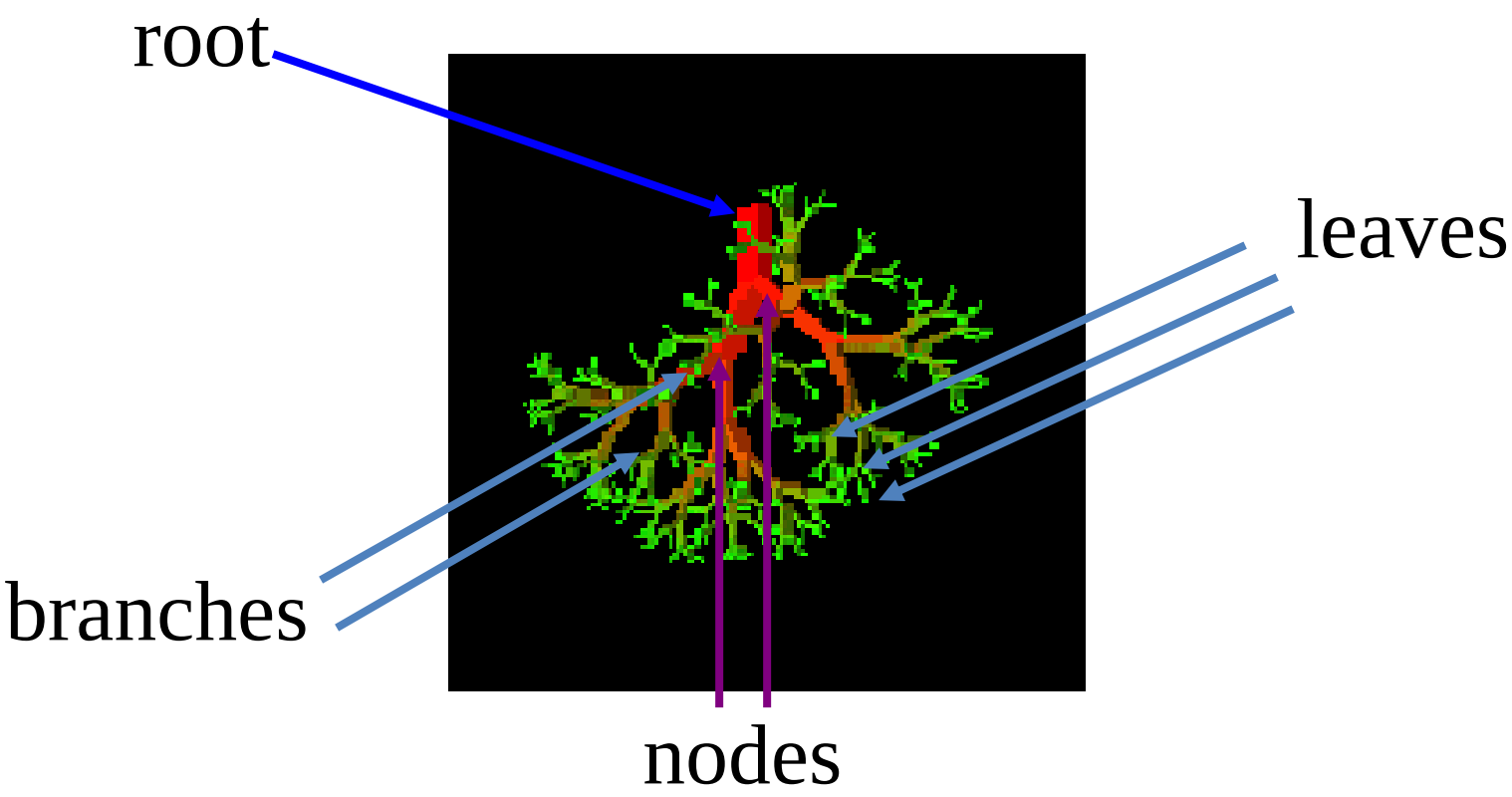
Basics of Tree

- Tree
  - Basic Terminologies
- Binary Tree
  - Binary Tree – Representation

Nature View of a Tree



Computer Scientist's View



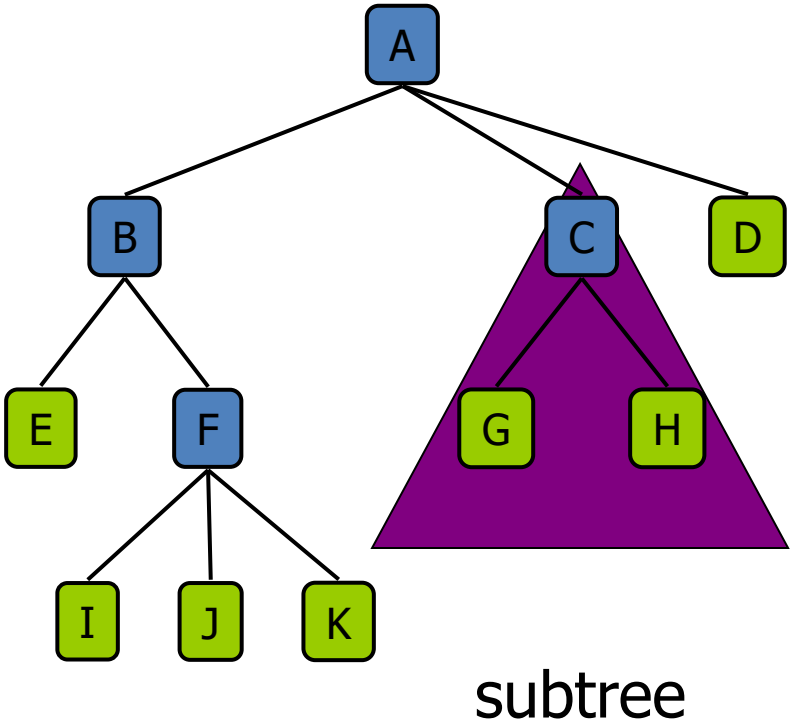
TREES

- Trees are one of the important non- Linear data structure.
- A tree is a Multilevel data structure that represent a hierarchical relationship between the Set of individual elements called nodes.
- Each tree structure starts with a node Which is called the root node of the Tree.

Tree Terminology

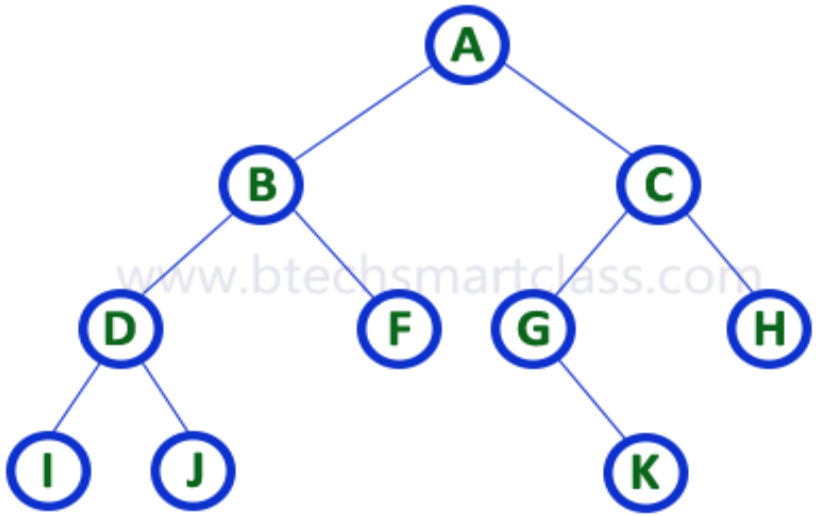
- **Root**: node without parent (A)
- **Siblings**: nodes share the same parent
- **Internal node**: node with at least one child (A, B, C, F)
- **External node** (leaf): node without children (E, I, J, K, G, H, D)
- **Ancestors** of a node: parent, grandparent, grand-grandparent, etc.
- **Descendant** of a node: child, grandchild, grand-grandchild, etc.
- **Depth** of a node: number of ancestors
- **Height** of a tree: maximum depth of any node (3)
- **Degree** of a node: the number of its children

- **Subtree**: tree consisting of a node and its descendants



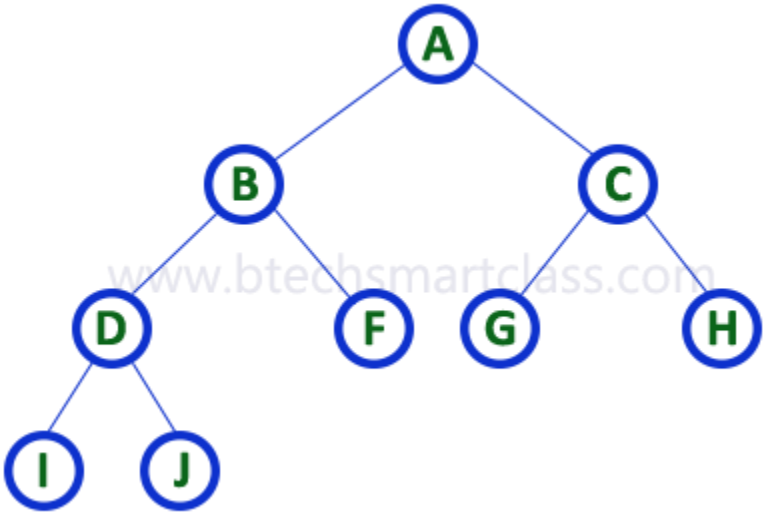
Binary Tree

- A tree in which every node can have a maximum of two children is called as Binary Tree.



Full Binary Tree

- A binary tree in which every node has either two or zero number of children is called Full Binary Tree



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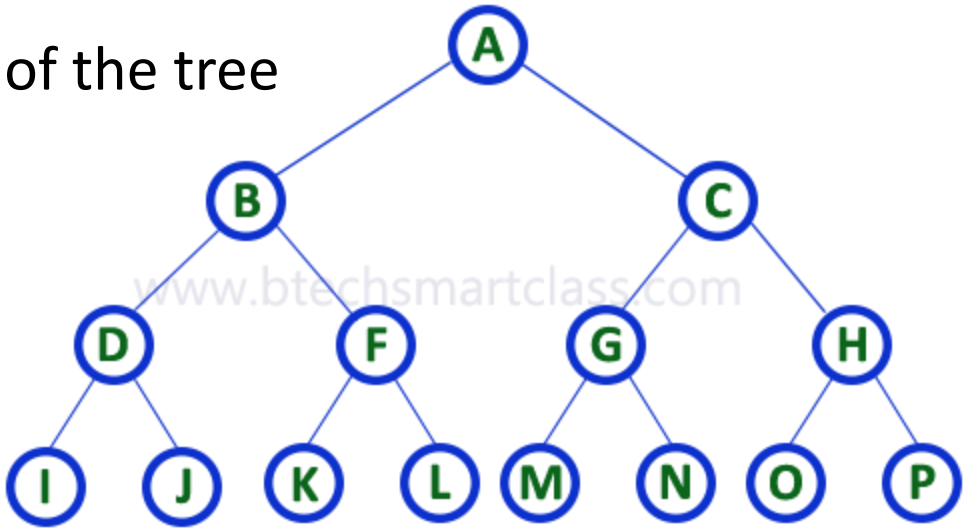
Complete Binary Tree

- A binary tree in which every internal node has exactly two children and all leaf nodes are possibly left at same level is called Complete Binary Tree. Complete binary tree is also called as **Perfect Binary Tree**
- **Number of nodes = $2^{d+1} - 1$**
- **Number of leaf nodes = 2^{d}**
- Where, d – Depth of the tree



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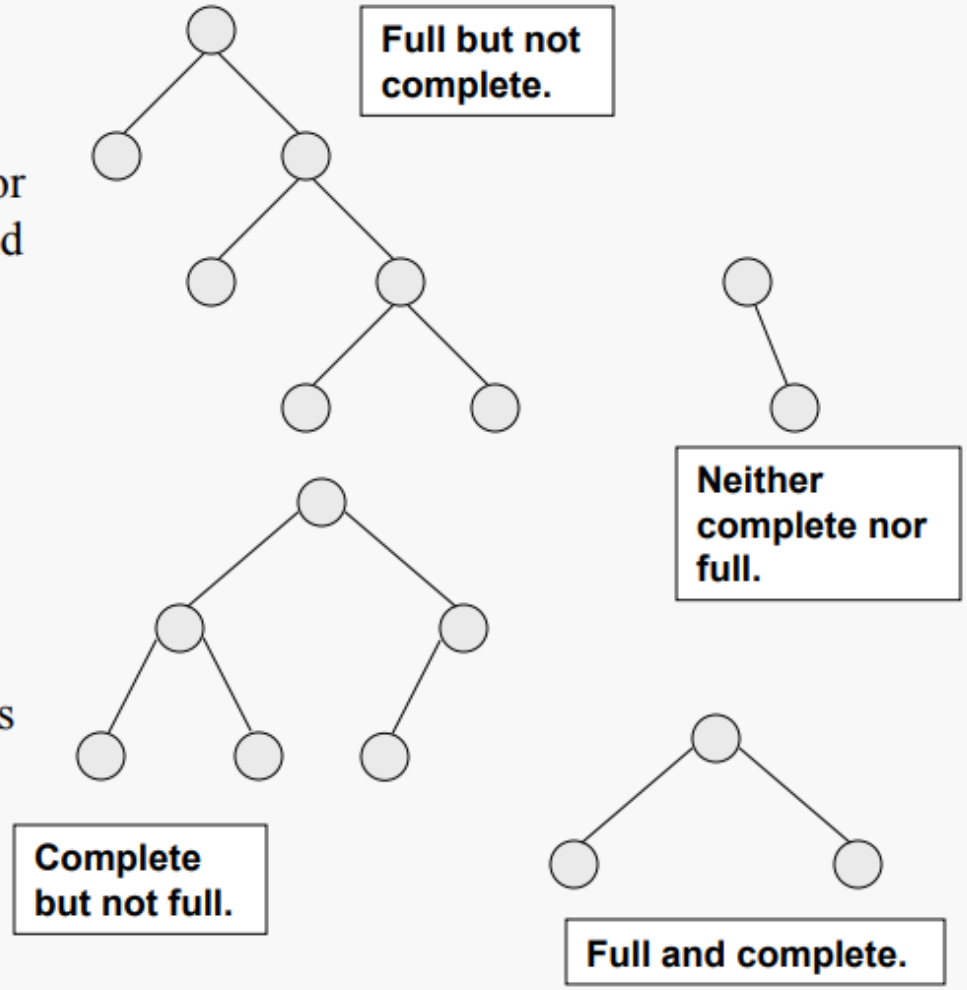
Complete vs. Full Binary Tree

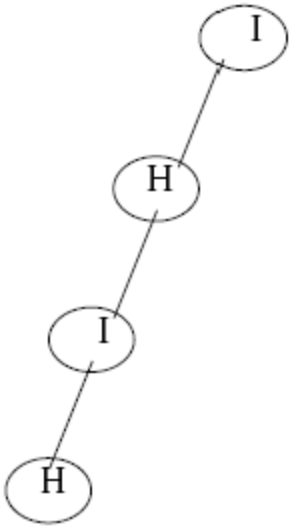
<u>Definition</u>: a binary tree T is *full* if each node is either a leaf or possesses exactly two child nodes.

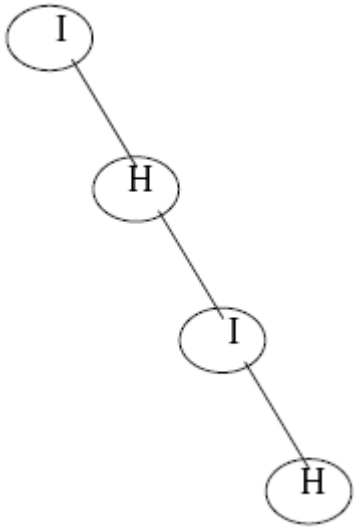
<u>Definition</u>: a binary tree T with n levels is *complete* if all levels except possibly the last are completely full, and the last level has all its nodes to the left side.



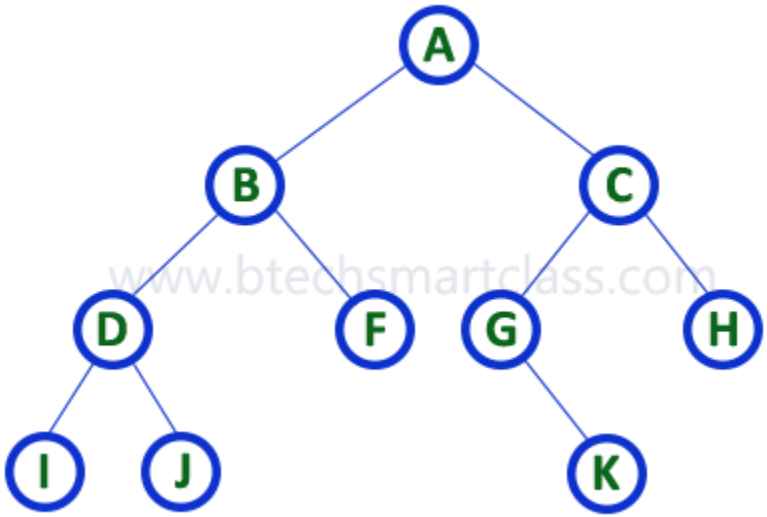
Left Skewed and Right Skewed Trees

- Binary tree has only left sub trees - Left Skewed Trees
- Binary tree has only right sub trees - Right Skewed Trees





Sequential representation



A	B	C	D	F	G	H	I	J	-	-	-	K	-	-
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

- To represent a binary tree of depth **'d'** using array representation, we need one dimensional array with a maximum size of $\mathbf{2^{d+1} - 1}$.

Sequential representation

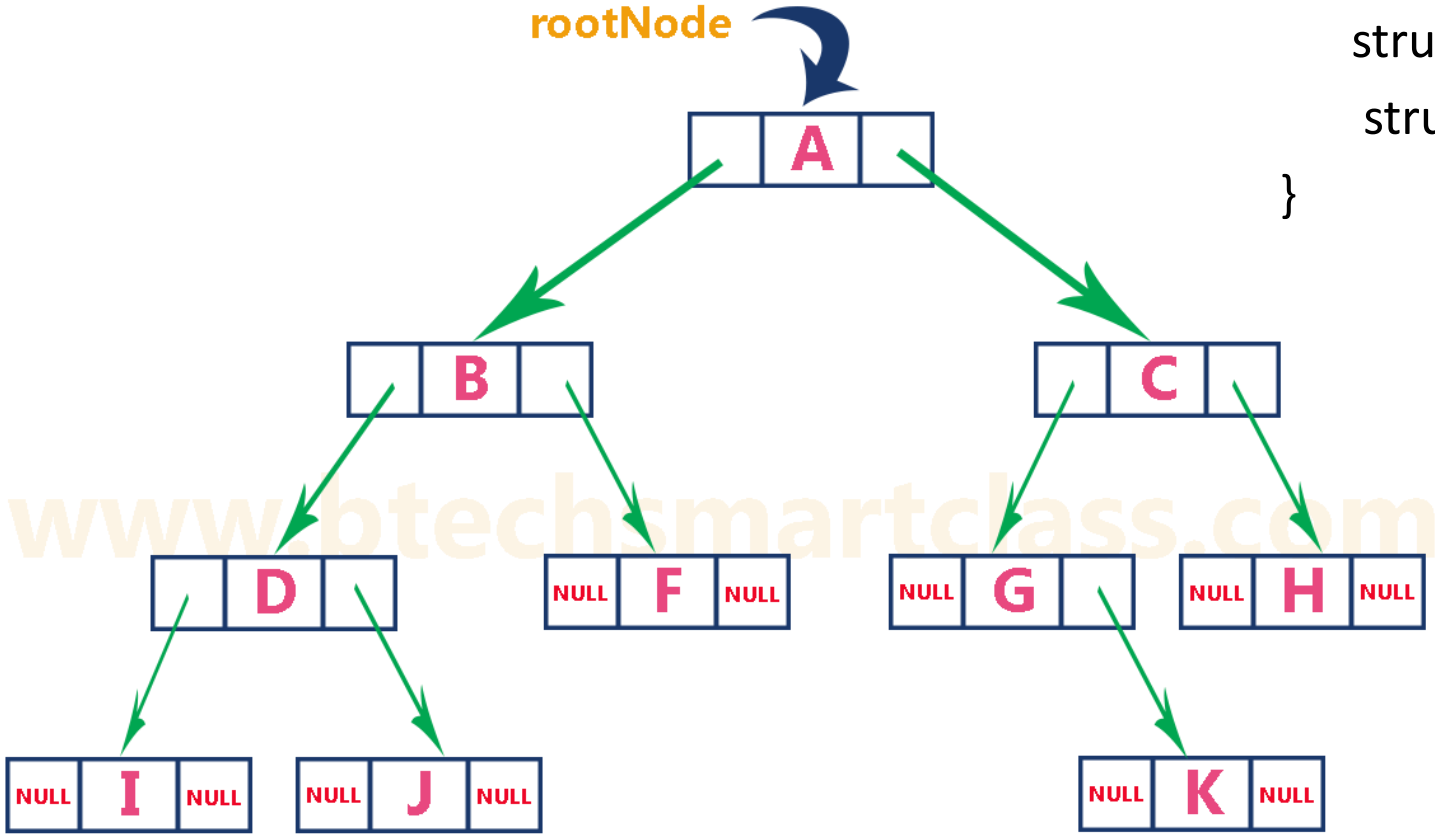
- Advantages:
  - Direct access to all nodes (Random access)
- Disadvantages:
  - Height of tree should be known
  - Memory may be wasted
  - Insertion and deletion of a node is difficult

Binary Tree Representation

1. Sequential representation using arrays
2. List representation using Linked list

List representation

```
struct node
{
    int data;
    struct node *left;
    struct node *rifgt;
}
```



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List representation

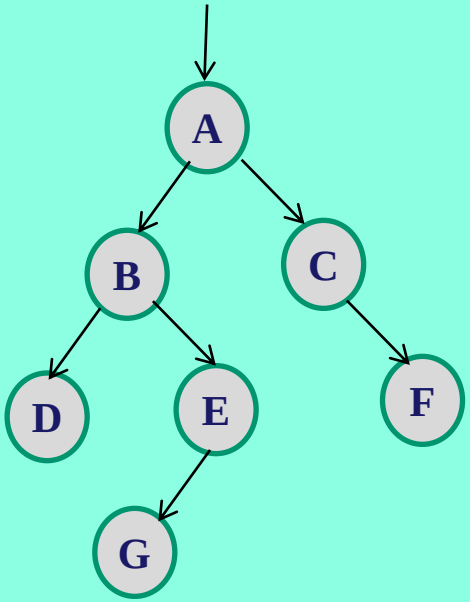
- Advantages:
  - Height of tree need not be known
  - No memory wastage
  - Insertion and deletion of a node is done without affecting other nodes
- Disadvantages:
  - Direct access to node is difficult
  - Additional memory required for storing address of left and right node

Non-Linear Data Structure

- In a non linear data structure , the Elements are not arranged in sequence.
- The data members are arranged in any Manner. The data items are not processed one After another.
- Trees and graphs are examples of non linear data structure.

Binary Tree

Topic 1: Create binary tree from a file



2 65 3

4 66 5

0 67 6

0 68 0

7 69 0

0 70 0

0 71 0

-1 -1 -1

Text File

```
#include <stdio.h>
struct node{
int data;
struct node *ll;
struct node *rl;
};

struct node *h,*q,*p,*ax[50];
int n;
void main(){
 void create();
 create();
return 0;
}
```

Binary Tree

Topic 1: Create binary tree from a file

```
#include <stdio.h>
struct node{
int data;
struct node *ll;
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};

struct node *h,*q,*p,*ax[50];
int n;

void main(){
 void create();
 create();
return 0;
}
```
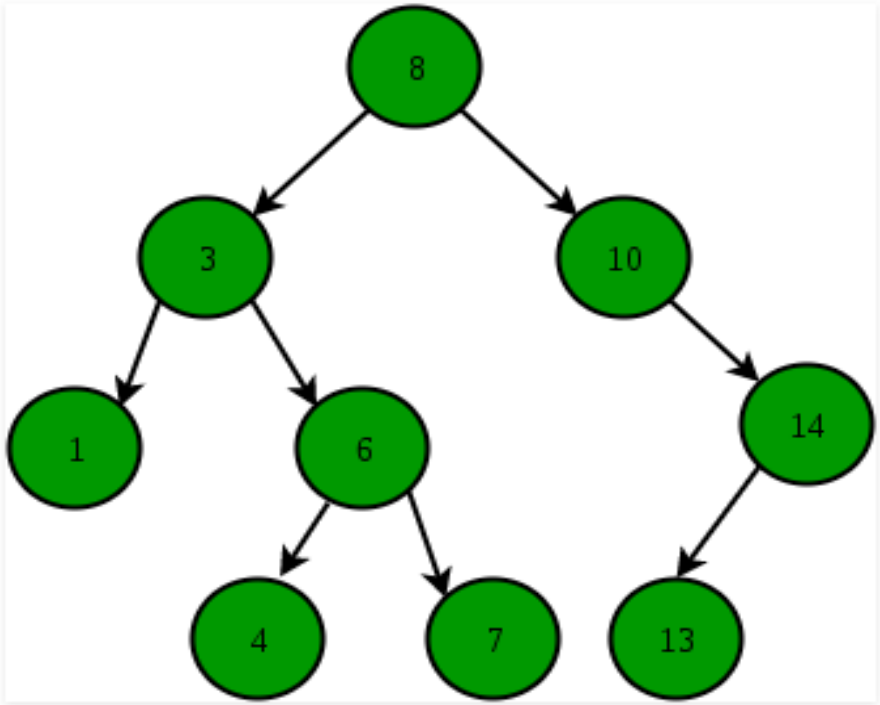

```
void create(){
FILE *fp;
fp=fopen("e:\\mycode\\mytree1.txt","r");
int i,x,y,z;

for(i=1;i<50;i++)
 ax[i]=(struct node*)(malloc(sizeof(struct node)));
ax[0]=0;

for(i=1;1<2;i++){
    fscanf(fp,"%d %d %d",&x,&y,&z);
    n=i;
    if(x==-1)break;
    ax[i]->data=y; ax[i]->ll=ax[x];ax[i]->rl=ax[z];
}
fclose(fp);
}
```

Binary Search Tree

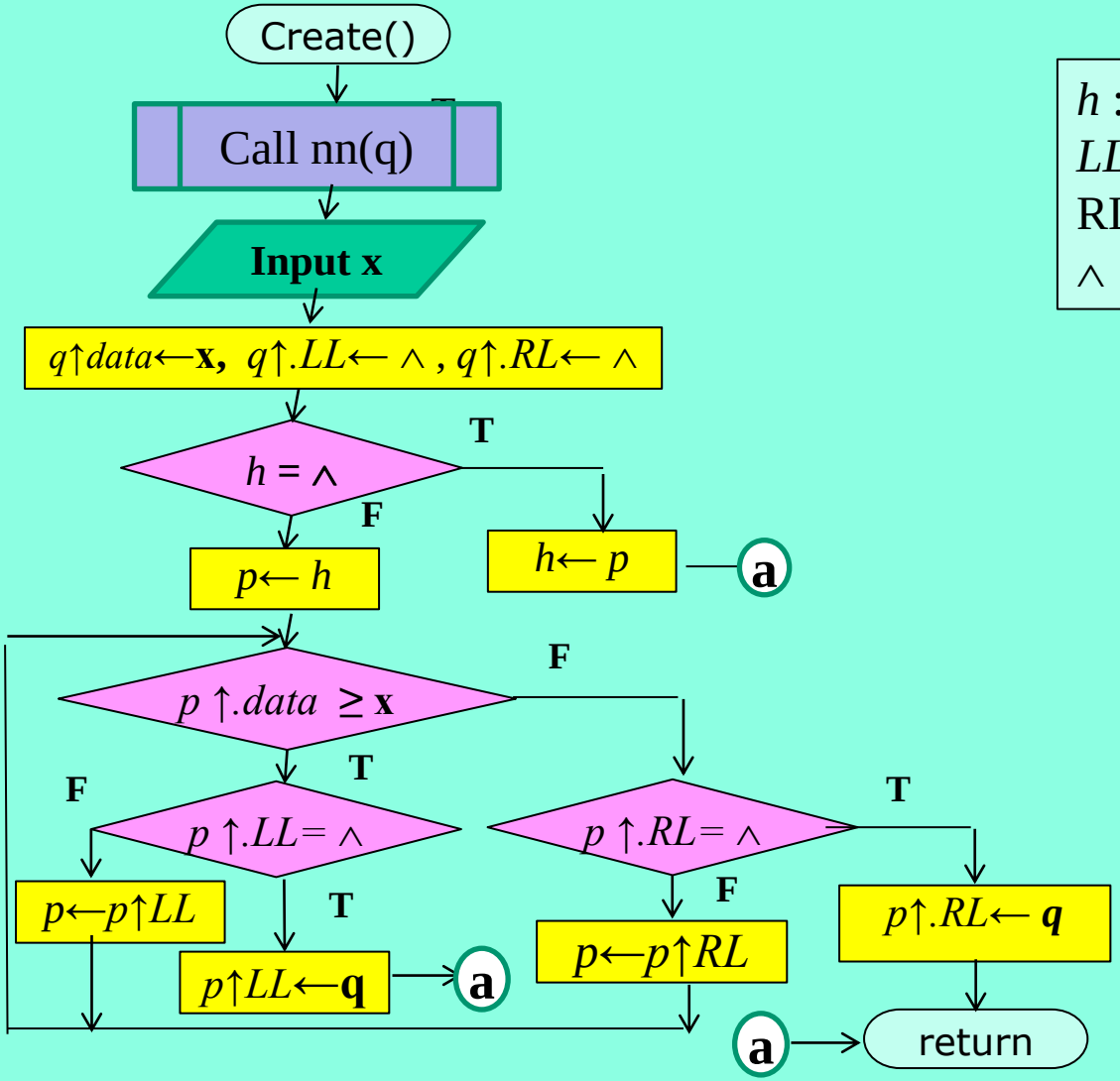
Binary Search Tree is a node-based binary tree data structure which has the following properties:

- The left subtree of a node contains only nodes with keys lesser than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- The left and right subtree each must also be a binary search tree.



Binary Search Tree

Topic 1: Write an Algorithm to insert a new node in BST



h : root address
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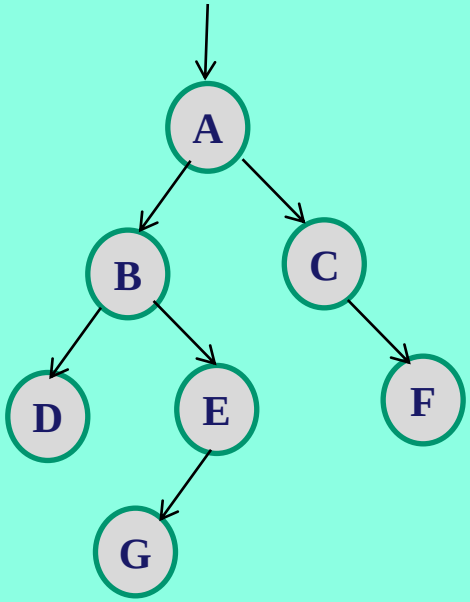
∧ : NULL



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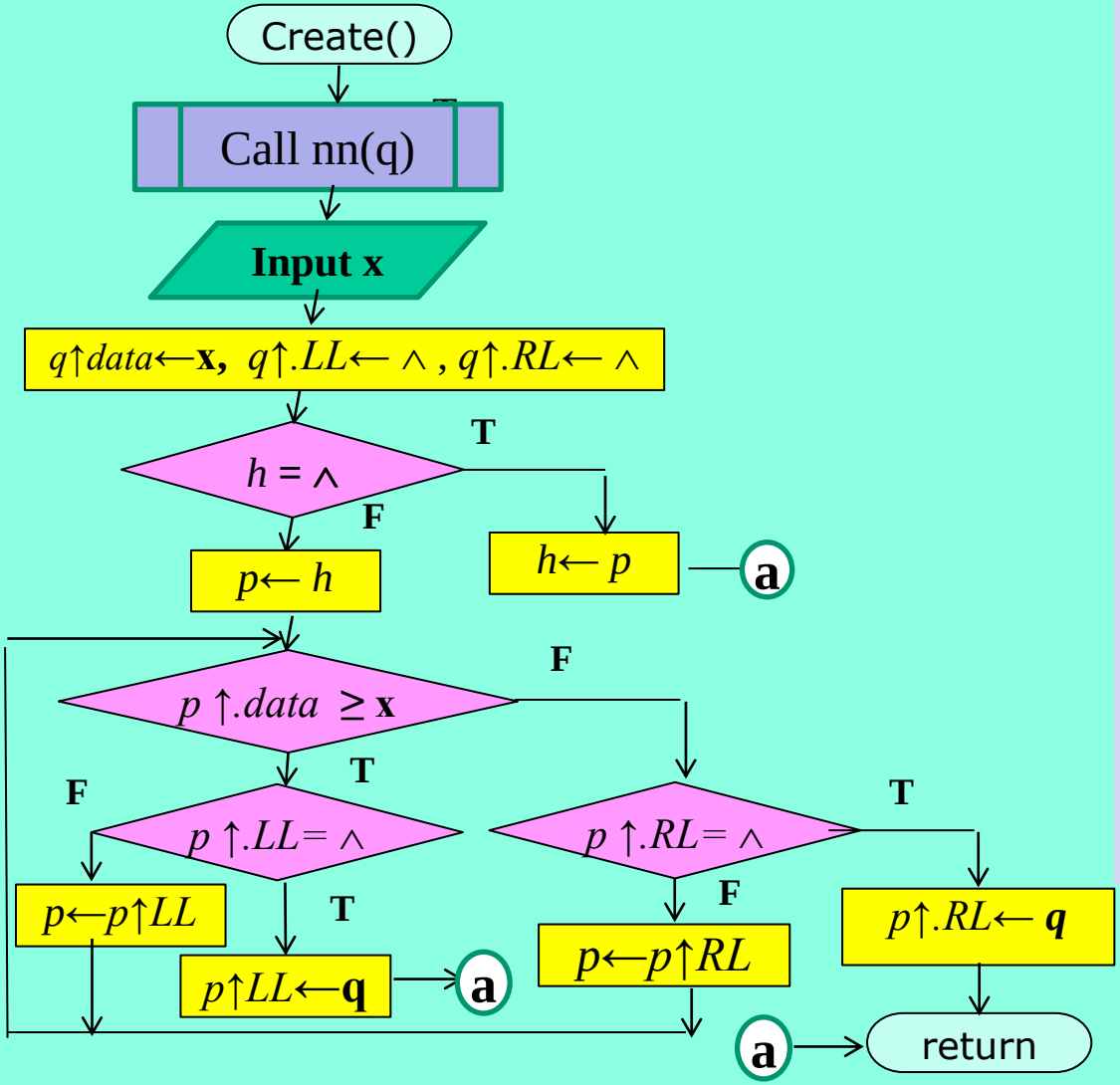
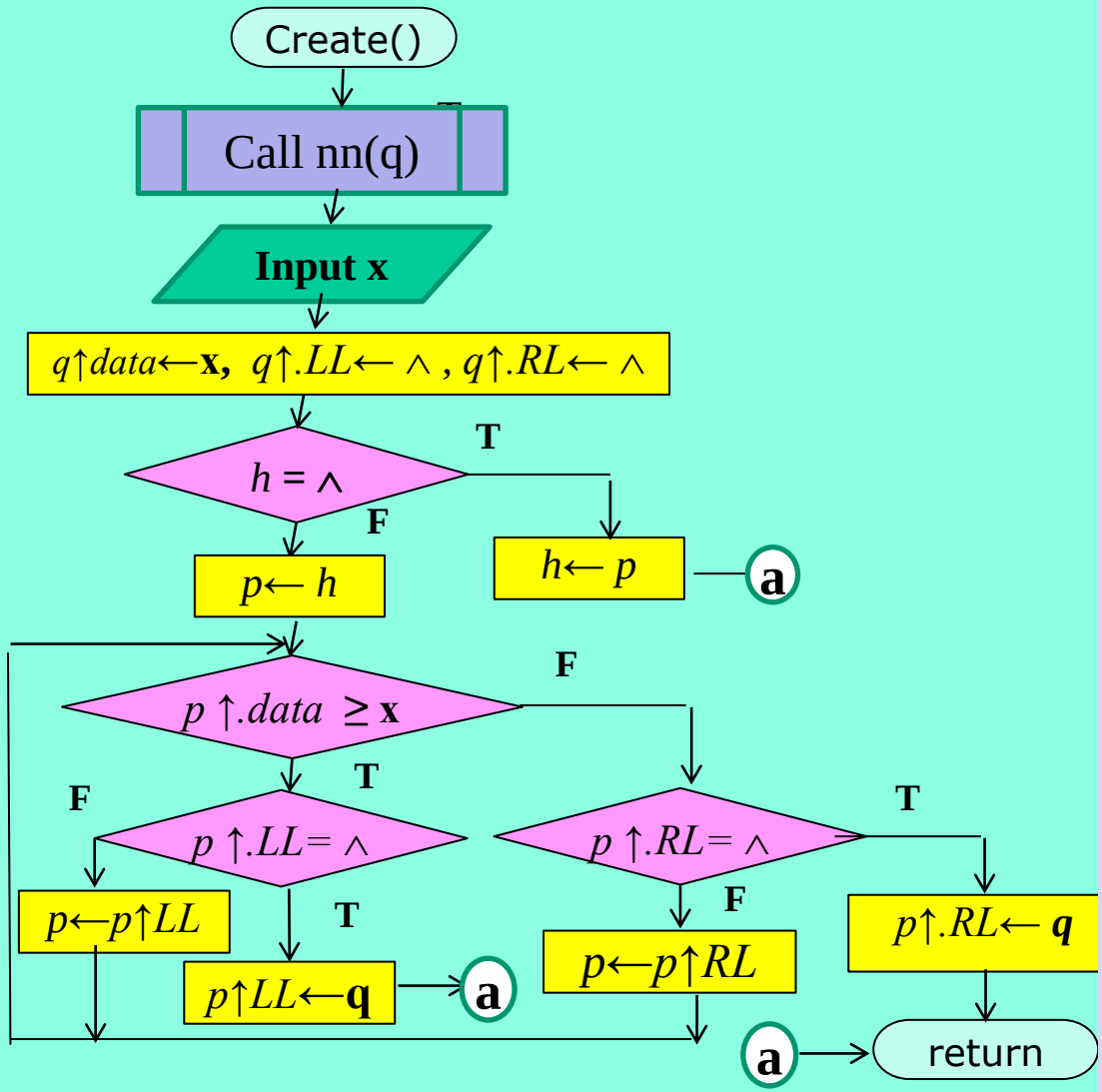



```
#include <stdio.h>
struct node{
int data;
struct node *ll;
struct node *rl;
};
struct node *h,*q,*p;
int n;
void main(){
 void insert();
 int i,x;
 h=0;
 for(;;){
    printf("Enter DATA:");
    scanf("%d",&x);
    if(x==0) break;
    insert(x);
 }
return 0;
}
```

Binary Search Tree

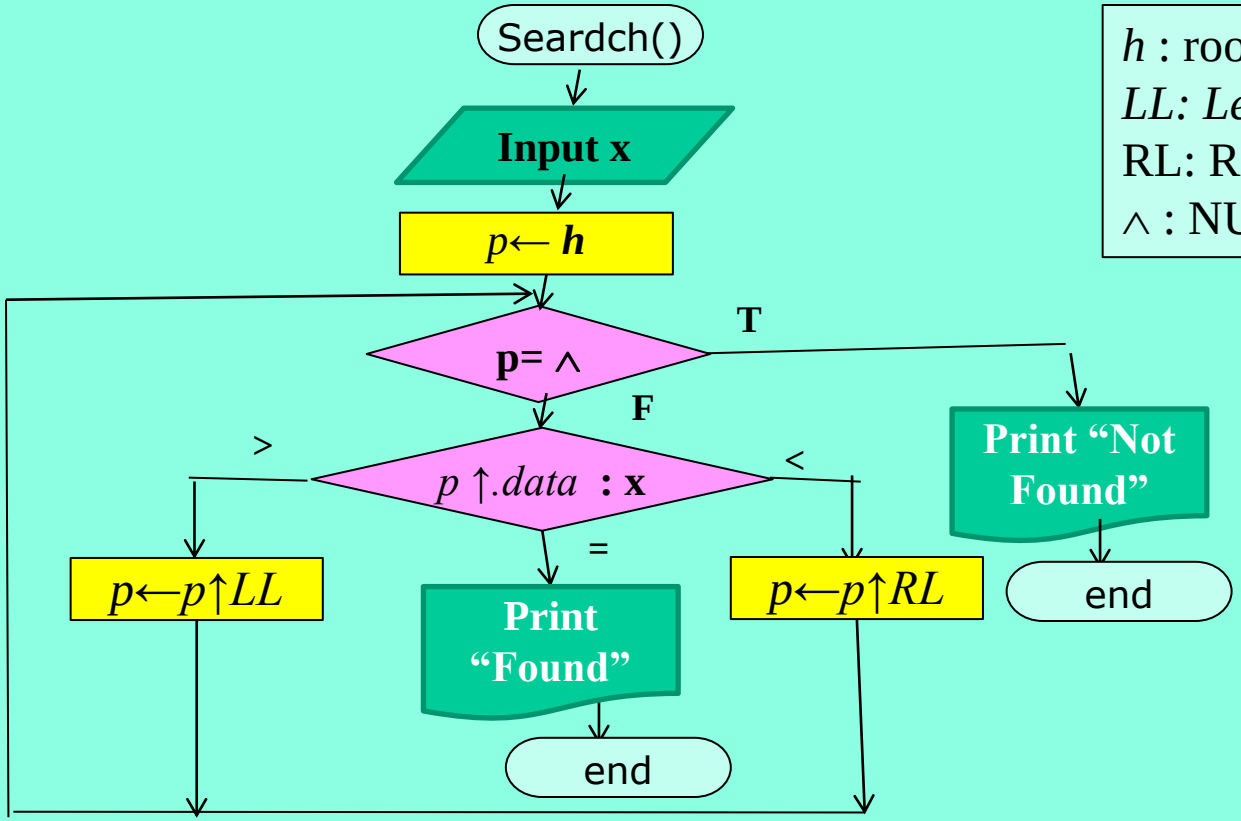
Topic 1: Write an Algorithm to insert a new node in BST.




```
void insert(int x){
  q=(struct node*)(malloc(sizeof(struct node)));
  printf("%X\n",q);
  q->data=x;q->ll=0;q->rl=0;
  if(h==0){  h=q; }
  else{
    p=h;
    for(;;){
    if(p->data>x)
      if(p->ll==0)
        {p->ll=q;
         printf("Node: %X %d %X\n",p->ll,p->data,p->rl);
         break;}
      else
        p=p->ll;
    else
        if(p->rl==0)
        {p->rl=q;
        printf("%X %d %X\n",p->ll,p->data,p->rl);
        break;}
      else
        p=p->rl;
    }
  }
}
```

Binary Search Tree

Topic 1: Write an Algorithm to Search a node in BST



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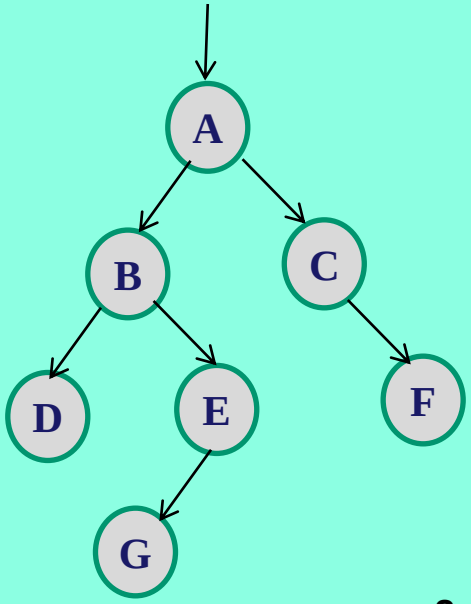


Binary Search Tree: (Delete)

When we delete a node, three possibilities arise.

1) ***Node to be deleted is leaf:*** Simply remove from the tree.

```
              50                            50
           /     \         delete(20)     /   \
          30      70       --------->    30     70
         /  \    /  \                     \    /  \
       20   40  60   80                    40  60   80
```

2) ***Node to be deleted has only one child:*** Copy the child to the node and delete the child

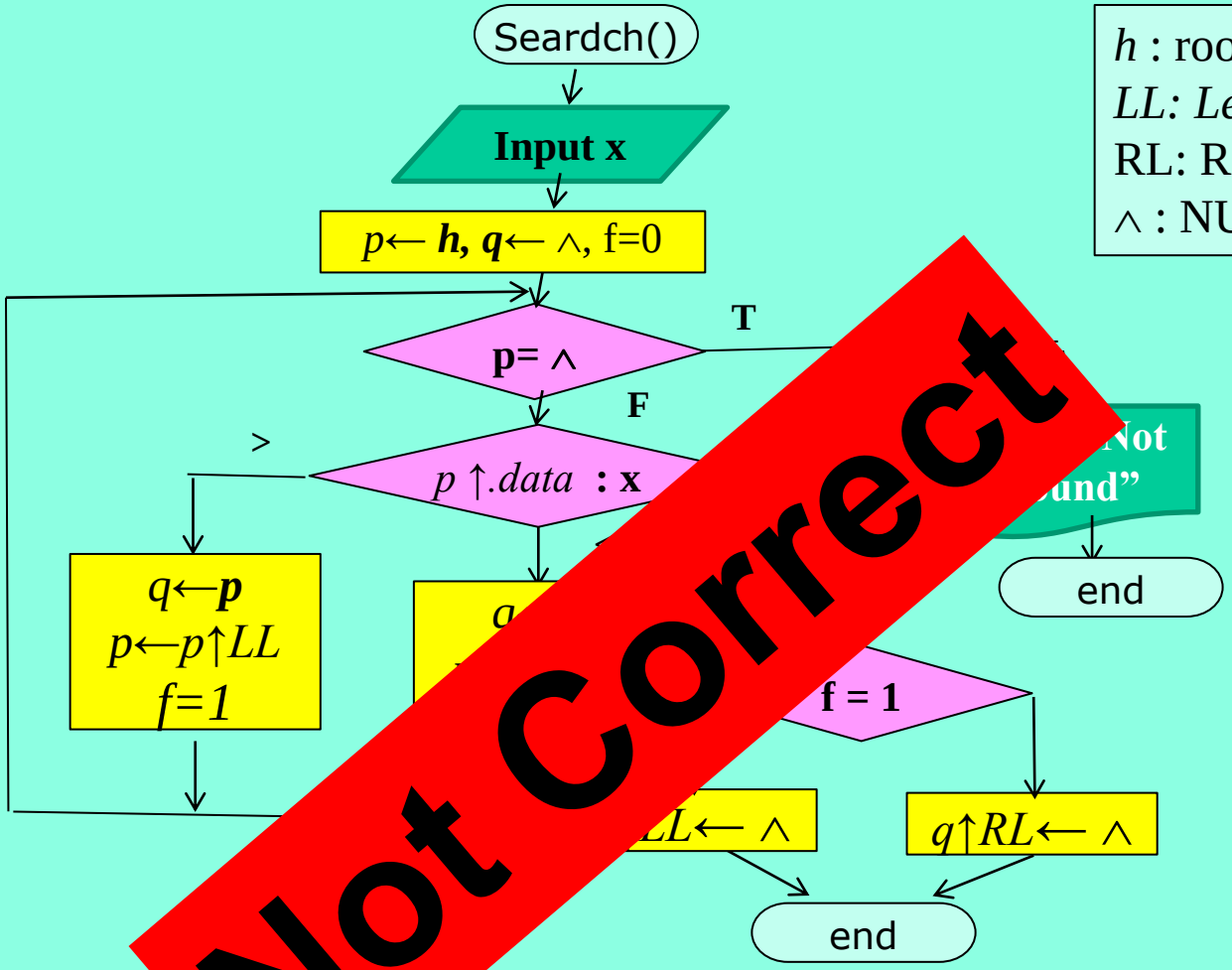
```
              50                            50
           /     \         delete(30)     /   \
          30      70       --------->    40     70
            \    /  \                           /  \
            40  60   80                        60   80
```

3) ***Node to be deleted has two children:*** Find inorder successor of the node. Copy contents of the inorder successor to the node and delete the inorder successor. Note that inorder predecessor can also be used.

```
              50                            60
           /     \         delete(50)     /   \
          40      70       --------->    40    70
                 /  \                            \
                60   80                           80
```

Binary Search Tree

Topic 1: Write an Algorithm to Delete a Leaf node in BST



Not Correct

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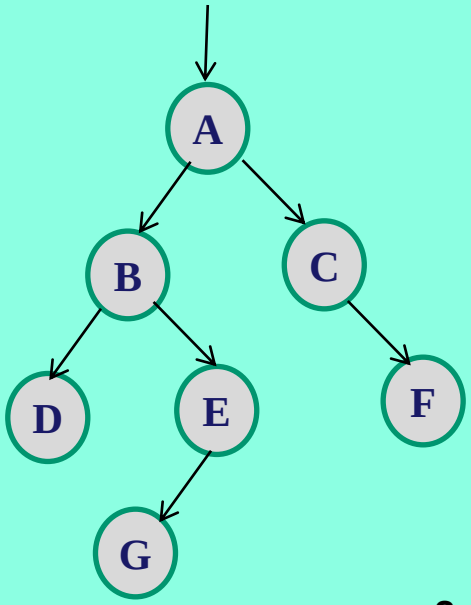
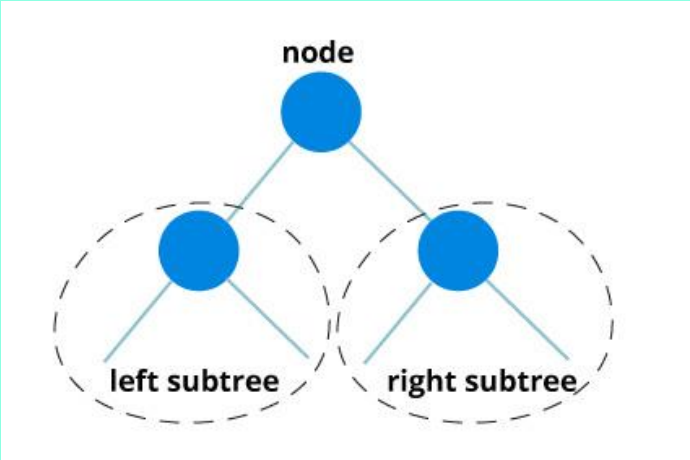


Tree Traversing

Traversing a tree means visiting every node in the tree. You might for instance want to add all the values in the tree or find the largest one. For all these operations, you will need to visit each node of the tree.



Inorder traversal (LDR)

1.First, visit all the nodes in the left subtree

2.Then the root node

3.Visit all the nodes in the right subtree

Preorder traversal (DLR)

1.Visit root node

2.Visit all the nodes in the left subtree

3.Visit all the nodes in the right subtree

Postorder traversal (LDR)

1.visit all the nodes in the left subtree

2.visit the root node

3.visit all the nodes in the right subtree

5 -> 12 -> 6 -> 1 -> 9

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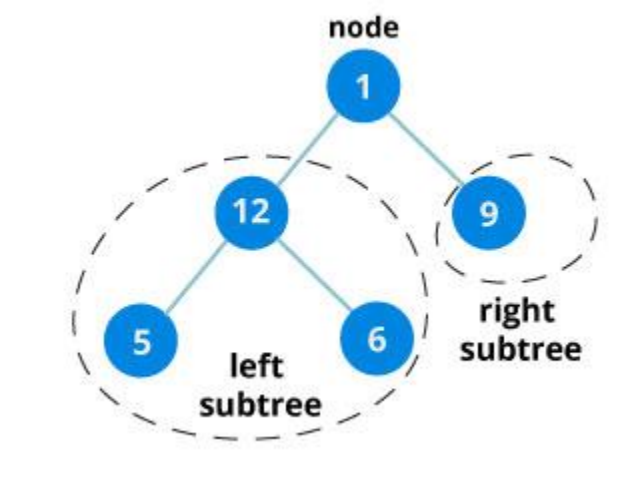

Inorder traversal

5 ->12 ->6 ->1 ->9

Preorder traversal

1 ->12 ->5 ->6 ->9

Postorder traversal

5 ->6 ->12 ->9 ->1

1. Inorder traversal (LDR)

1.First, visit all the nodes in the left subtree

2.Then the root node

3.Visit all the nodes in the right subtree

2. Preorder traversal (DLR)

1.Visit root node

2.Visit all the nodes in the left subtree

3.Visit all the nodes in the right subtree

3. Postorder traversal (LDR)

1.visit all the nodes in the left subtree

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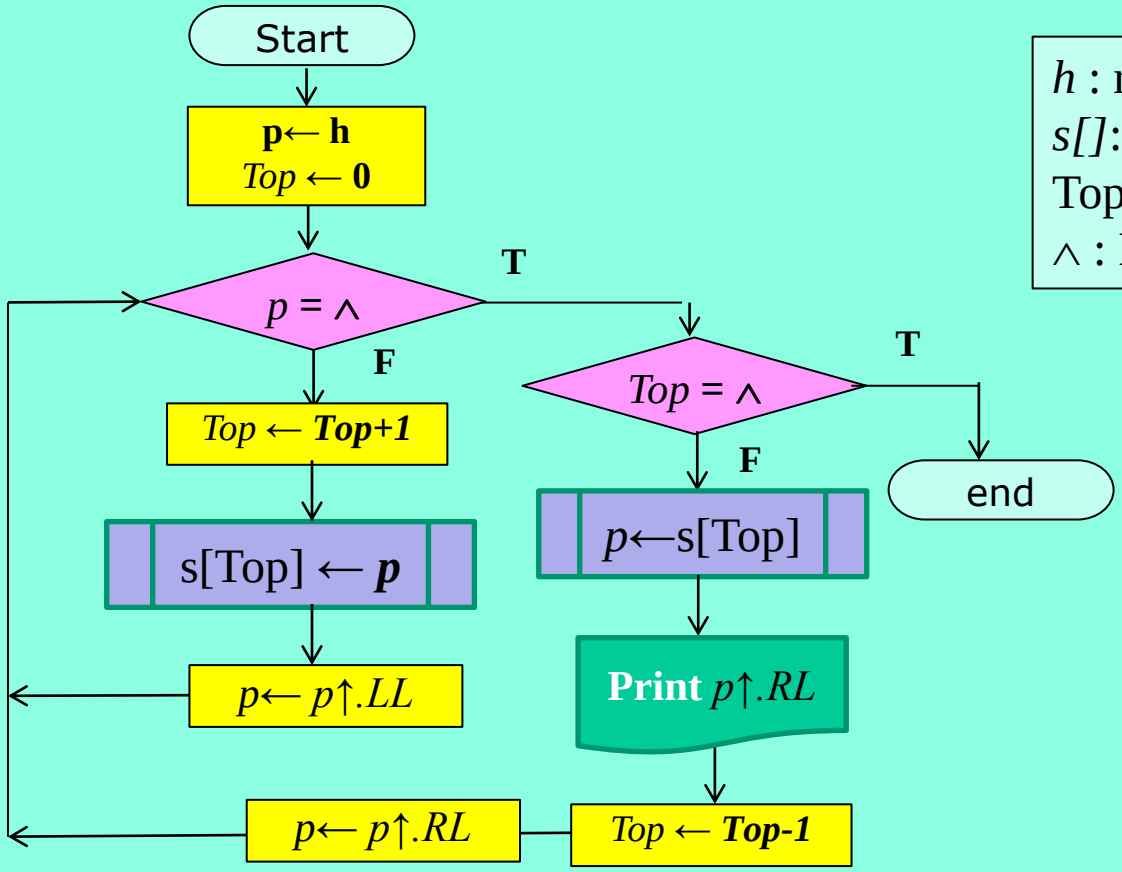
Traversing BT

Topic 1: Write an Algorithm to traverse a binary tree in inorder traversing.



h : root address
s[]: stack
Top : Stack Pointer
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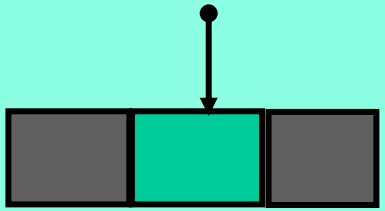
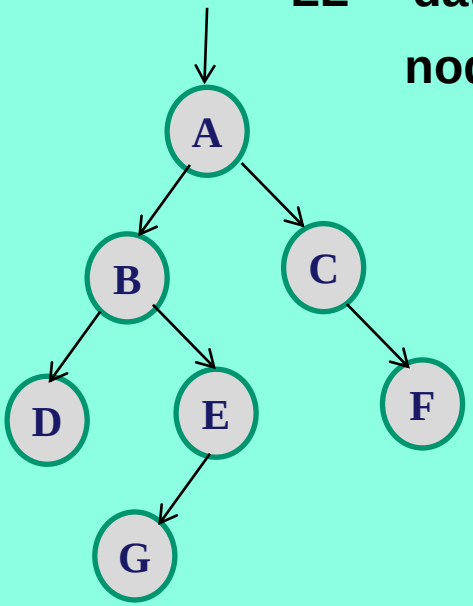


node



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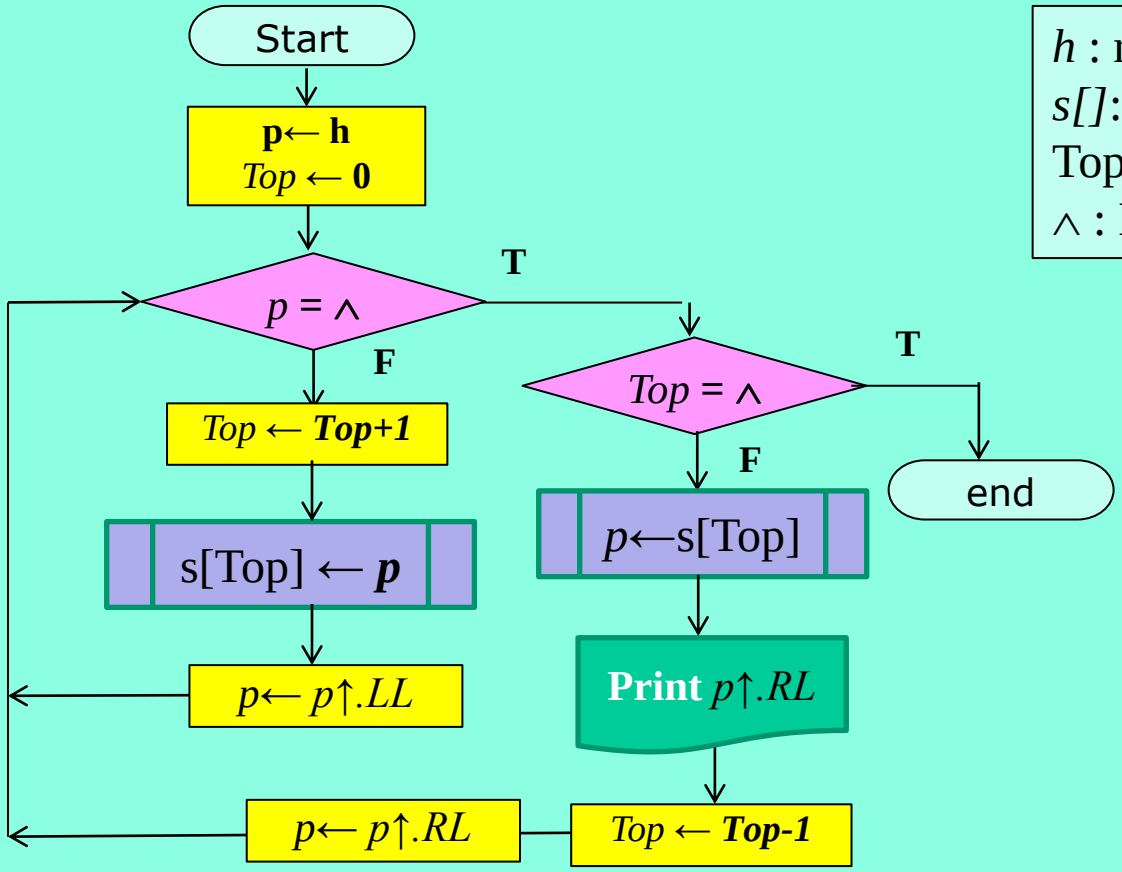

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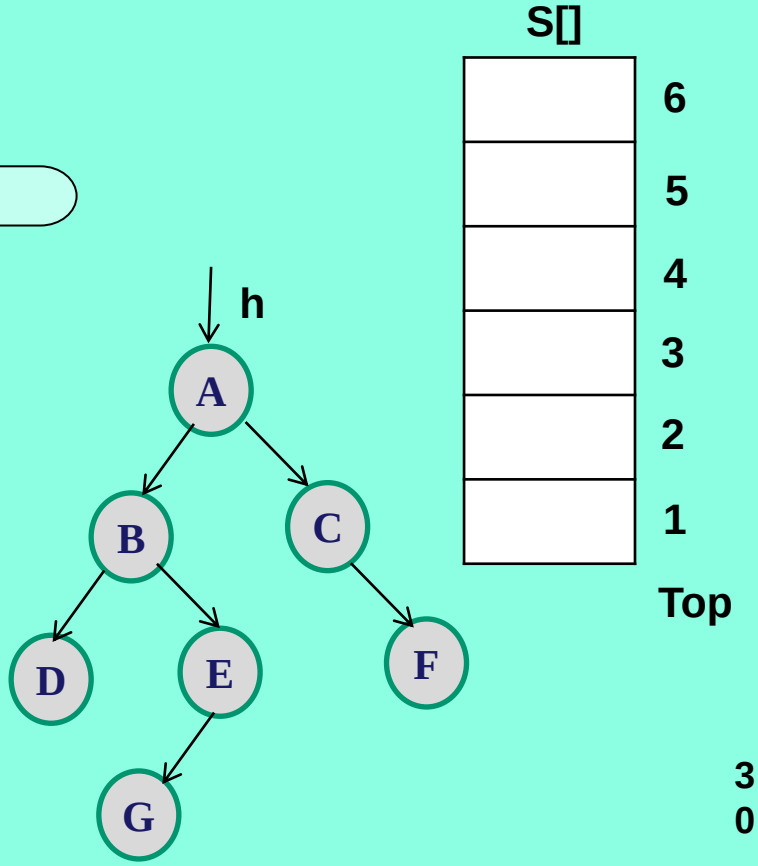

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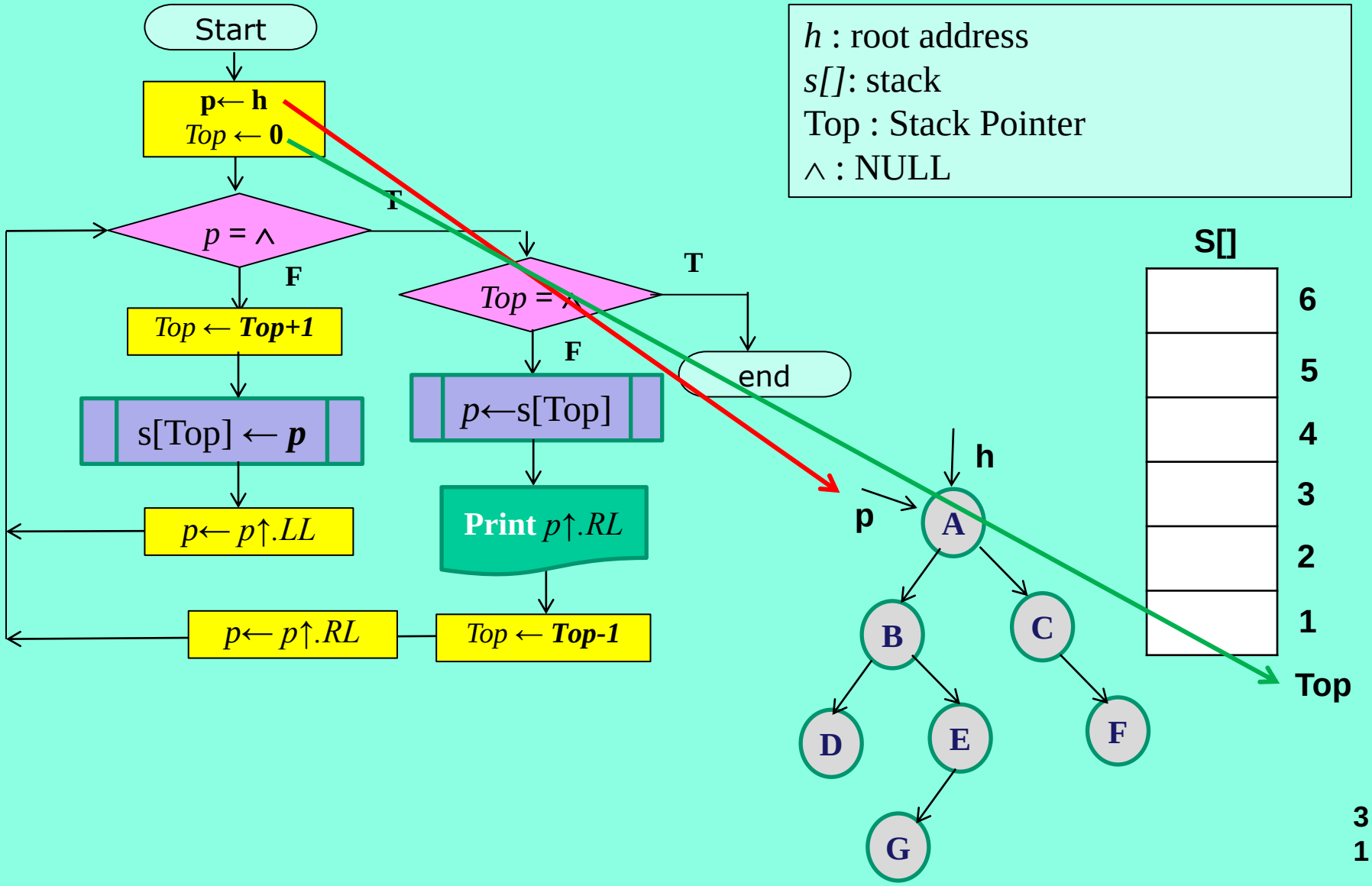

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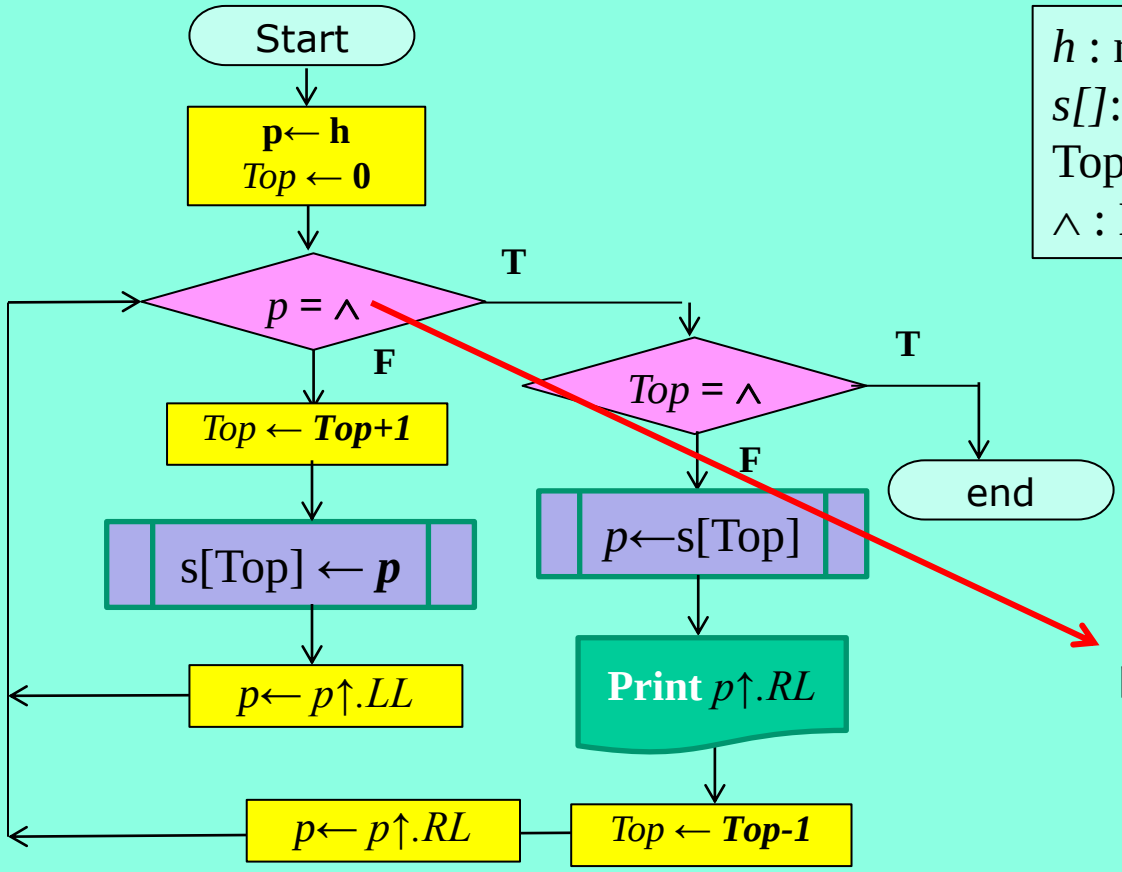

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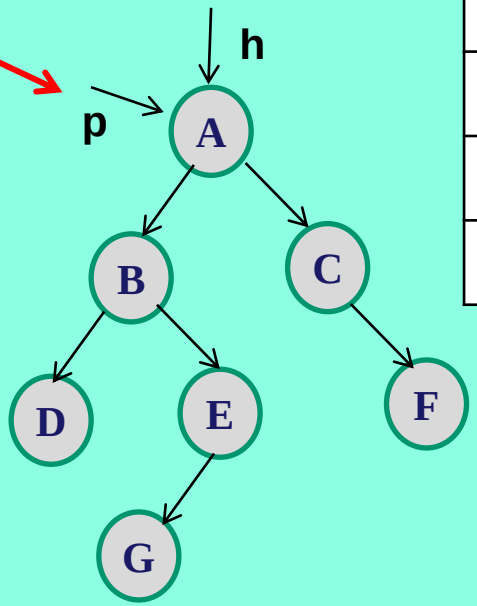

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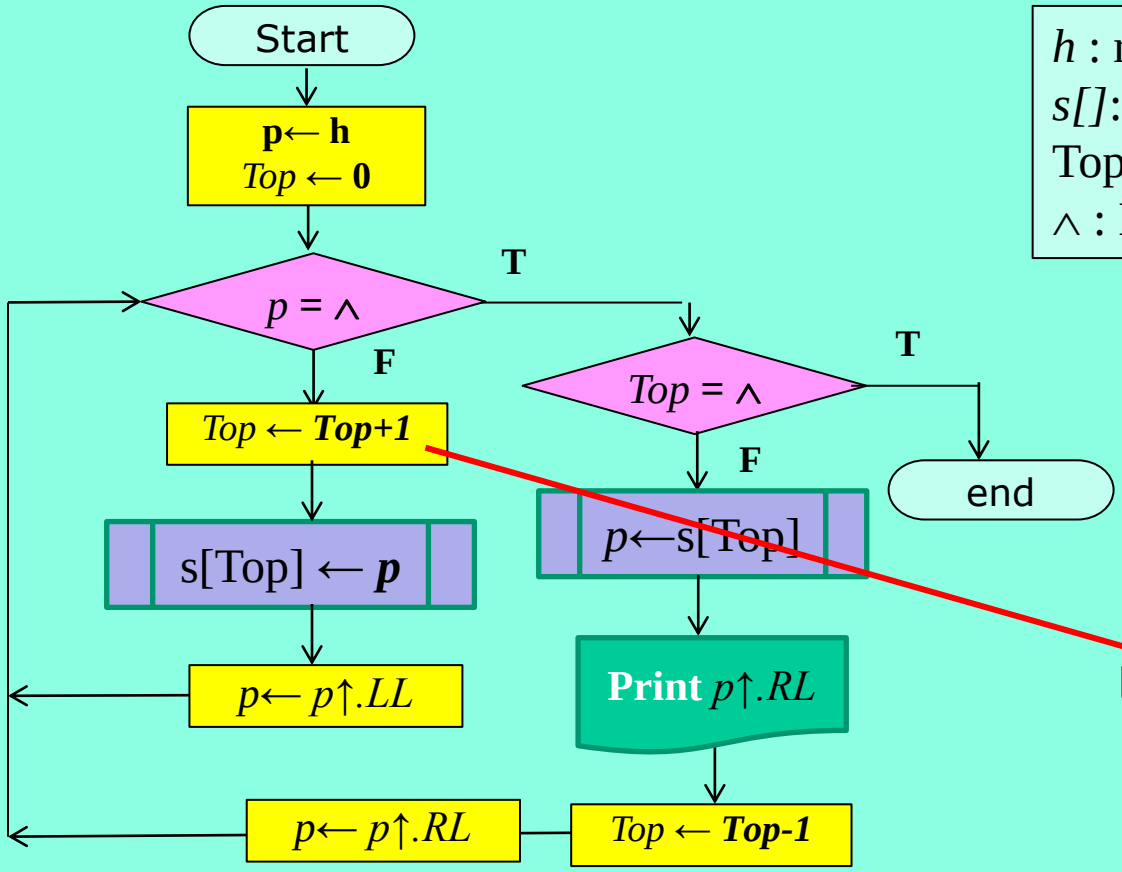

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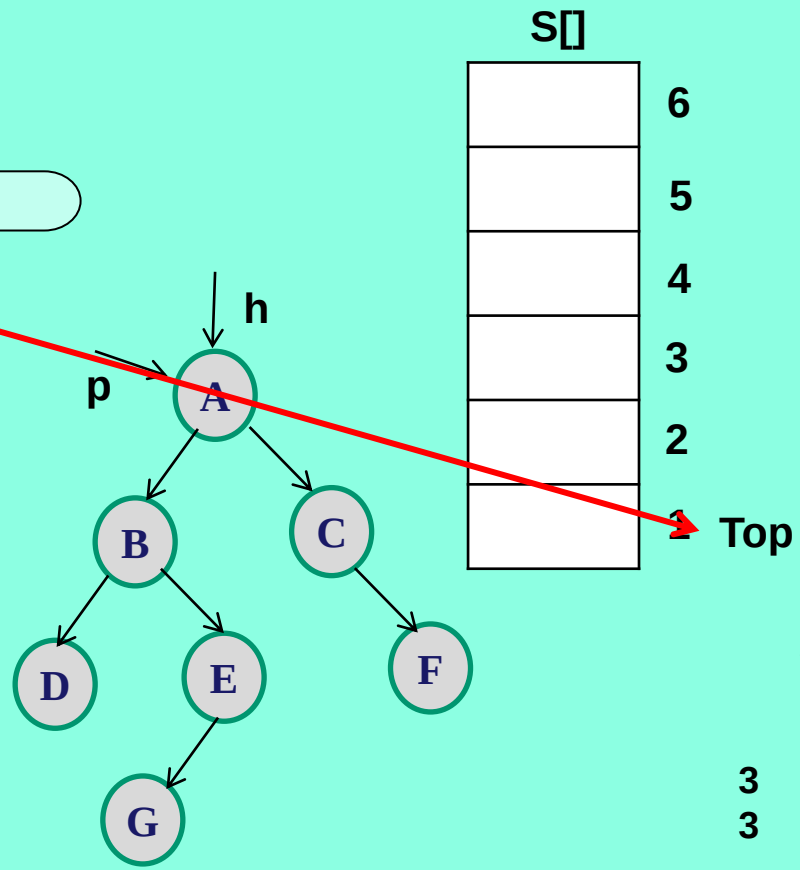

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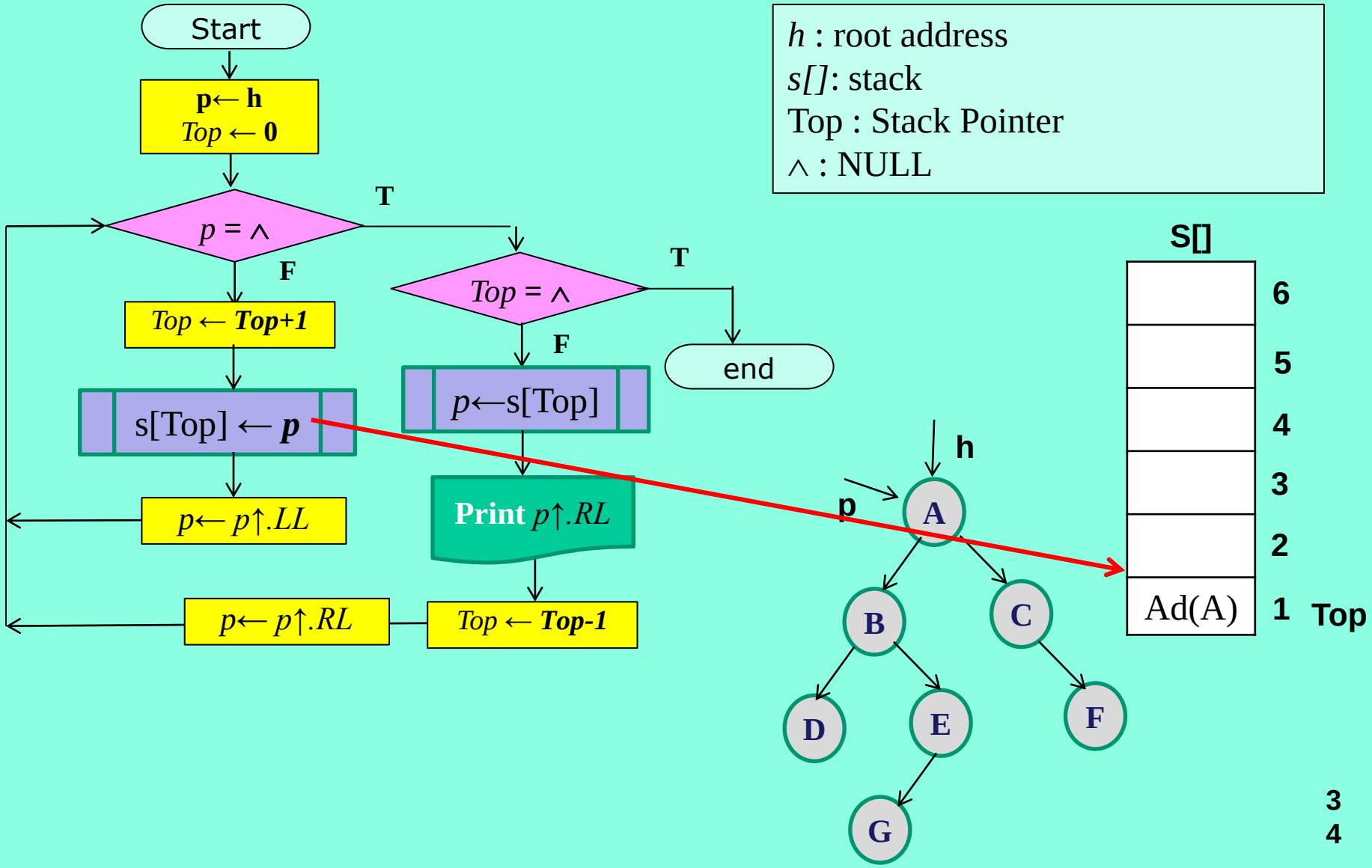

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Traversing BT

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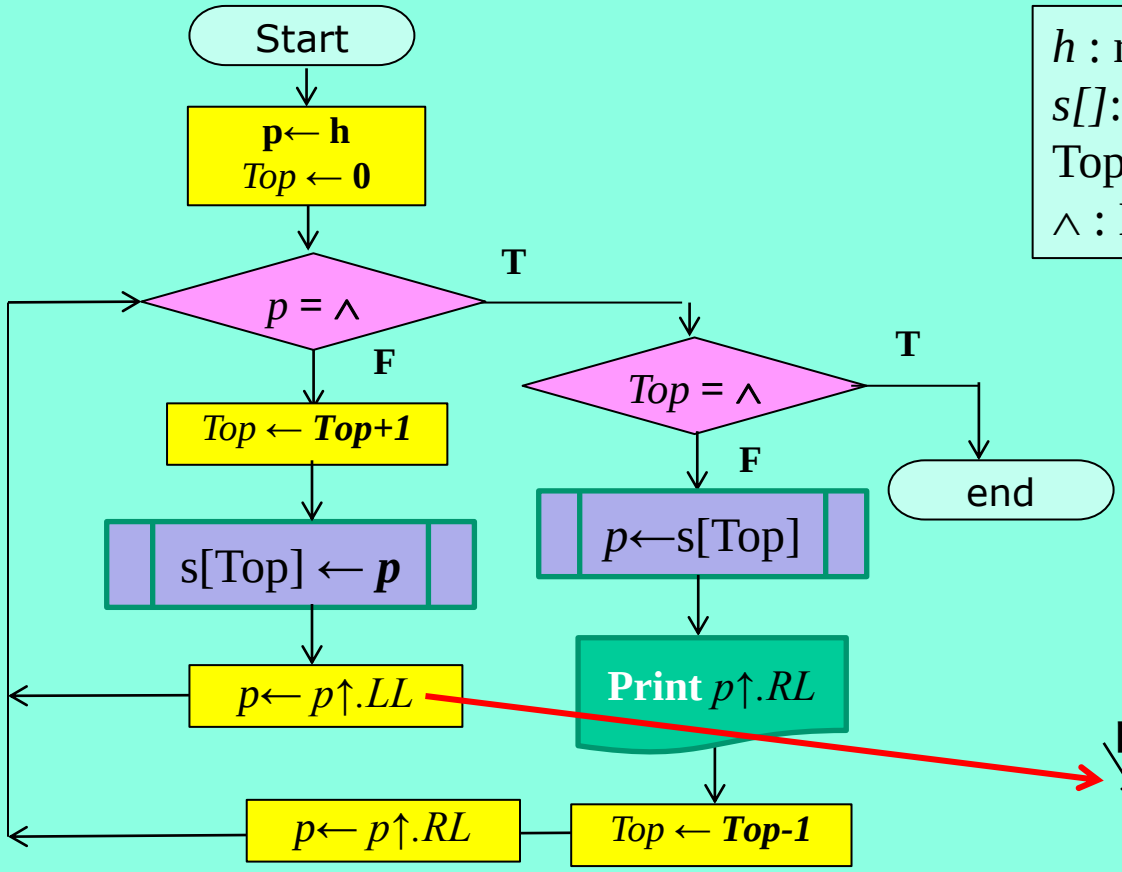

h : root address
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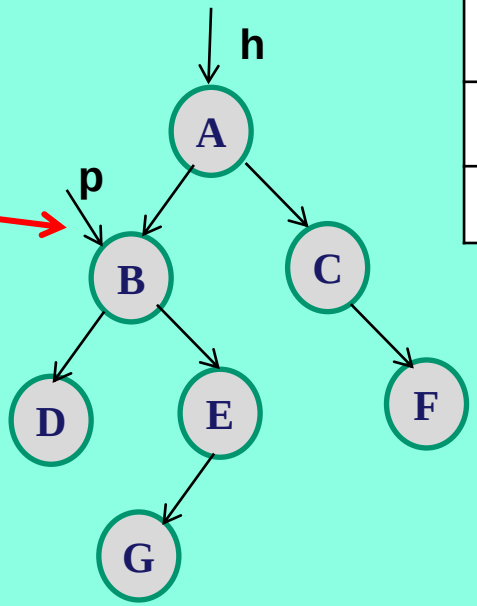

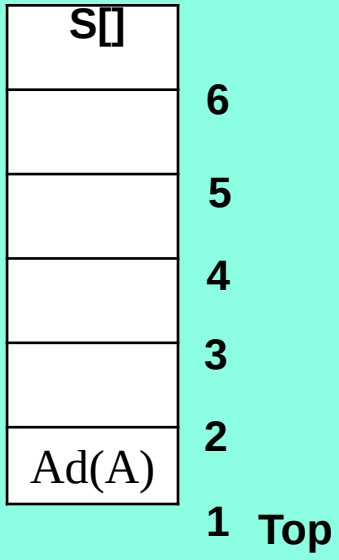

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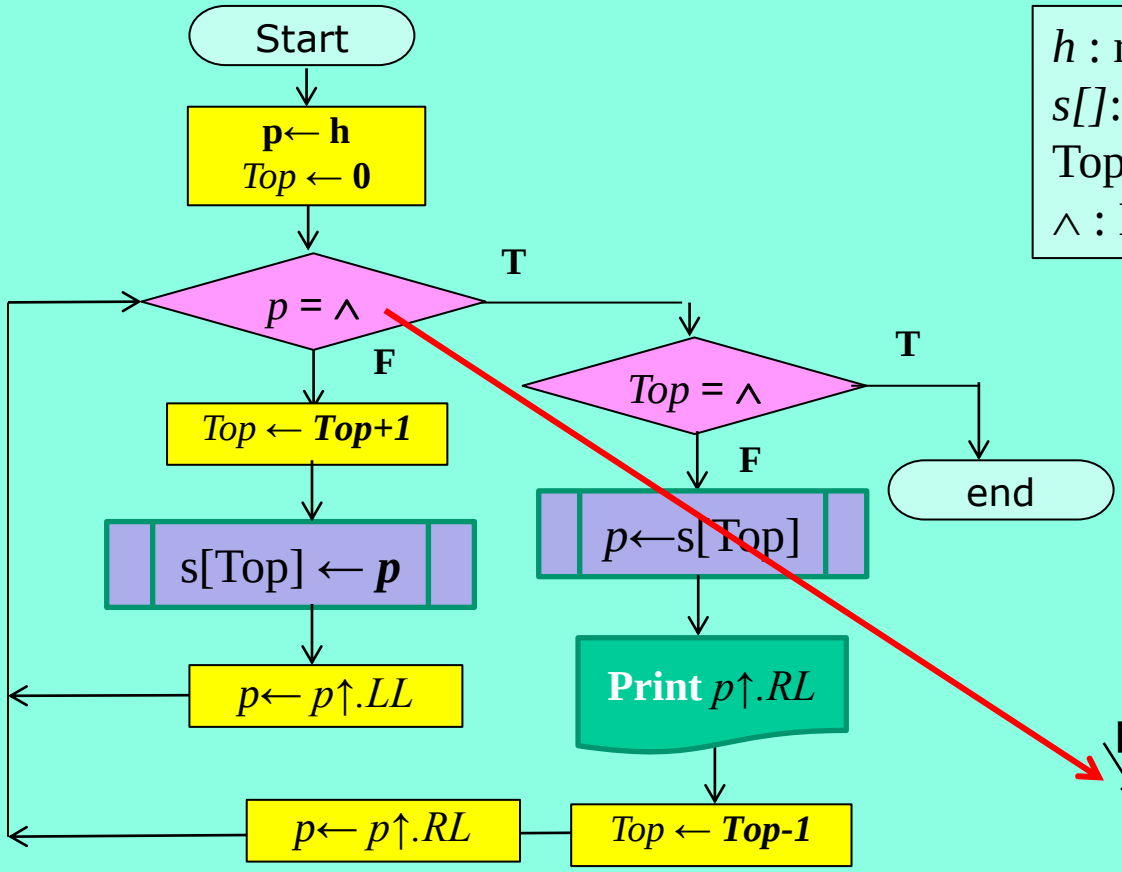

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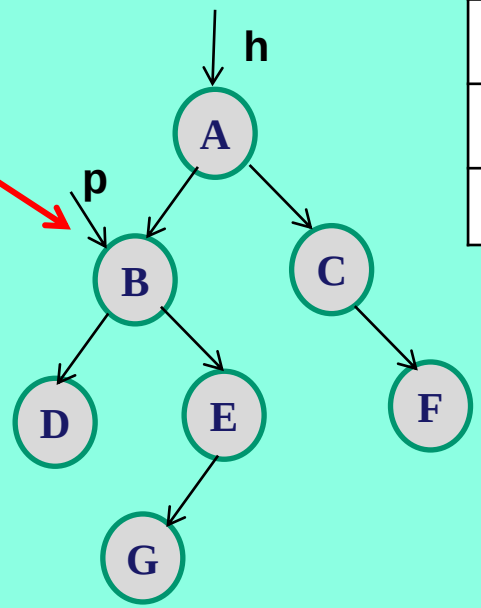

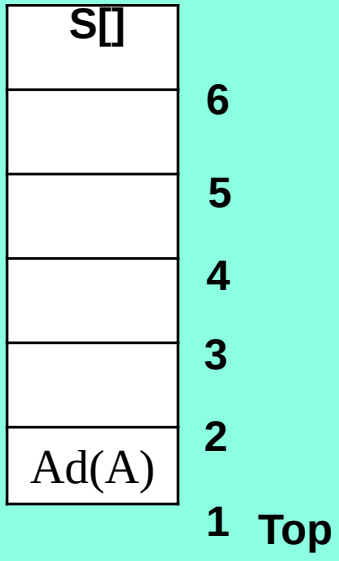

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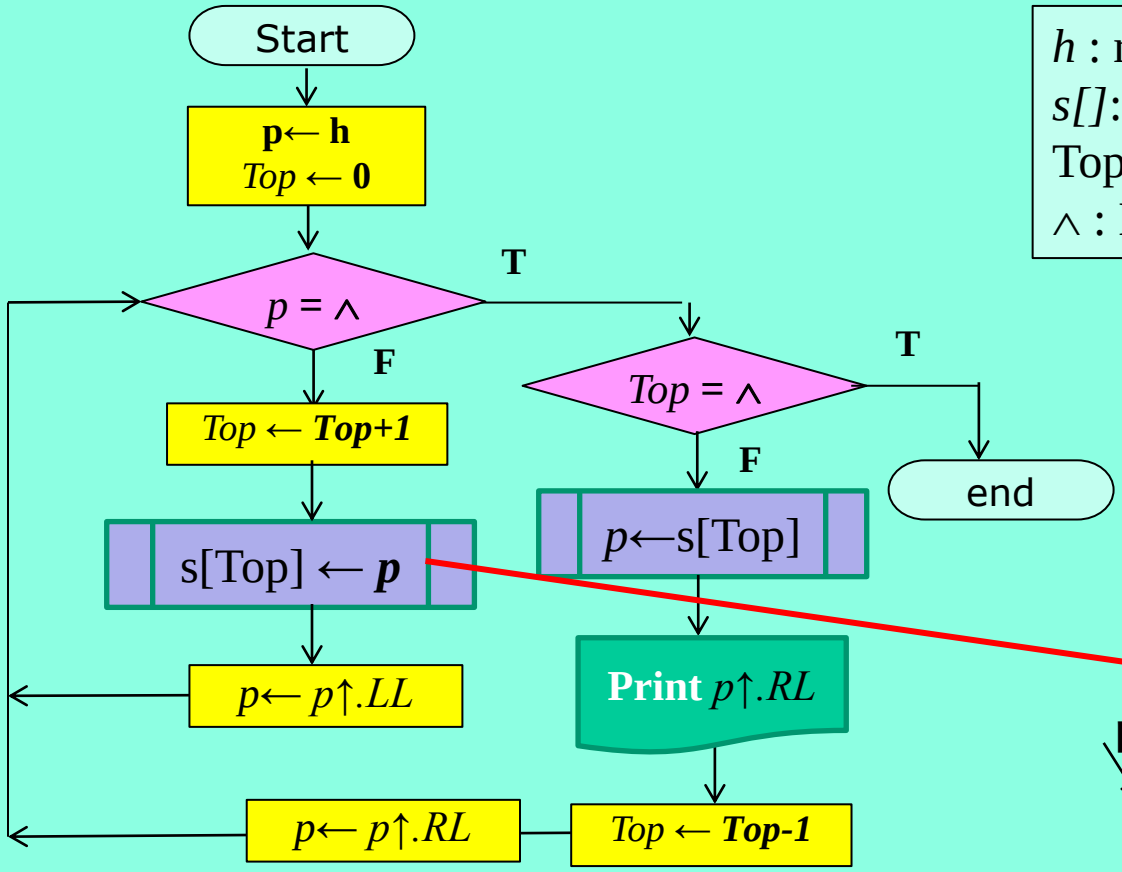

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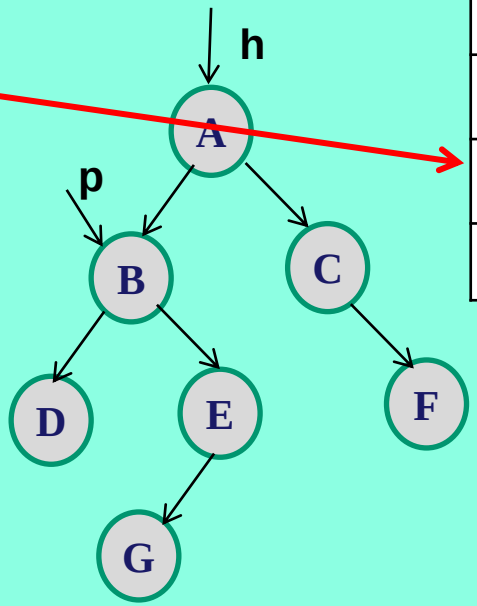

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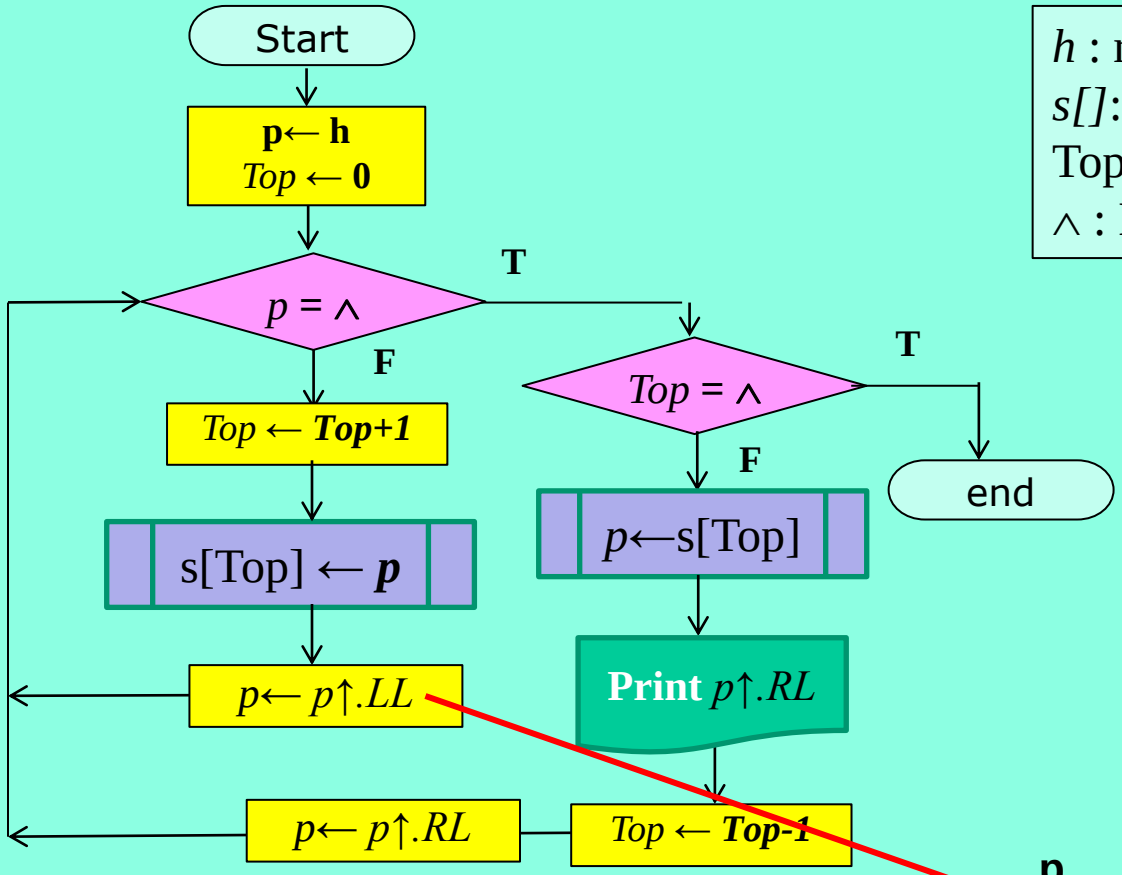

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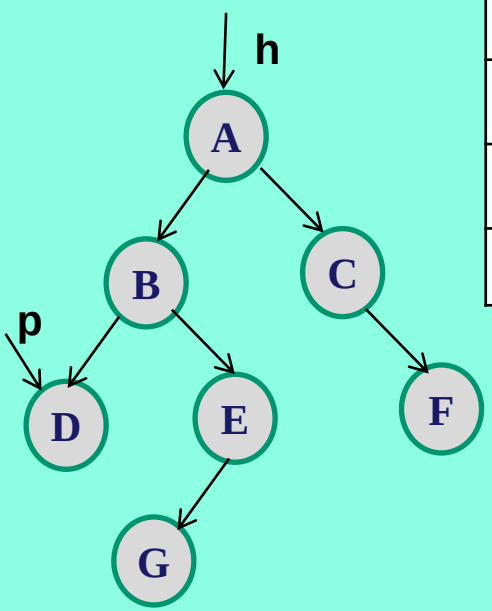

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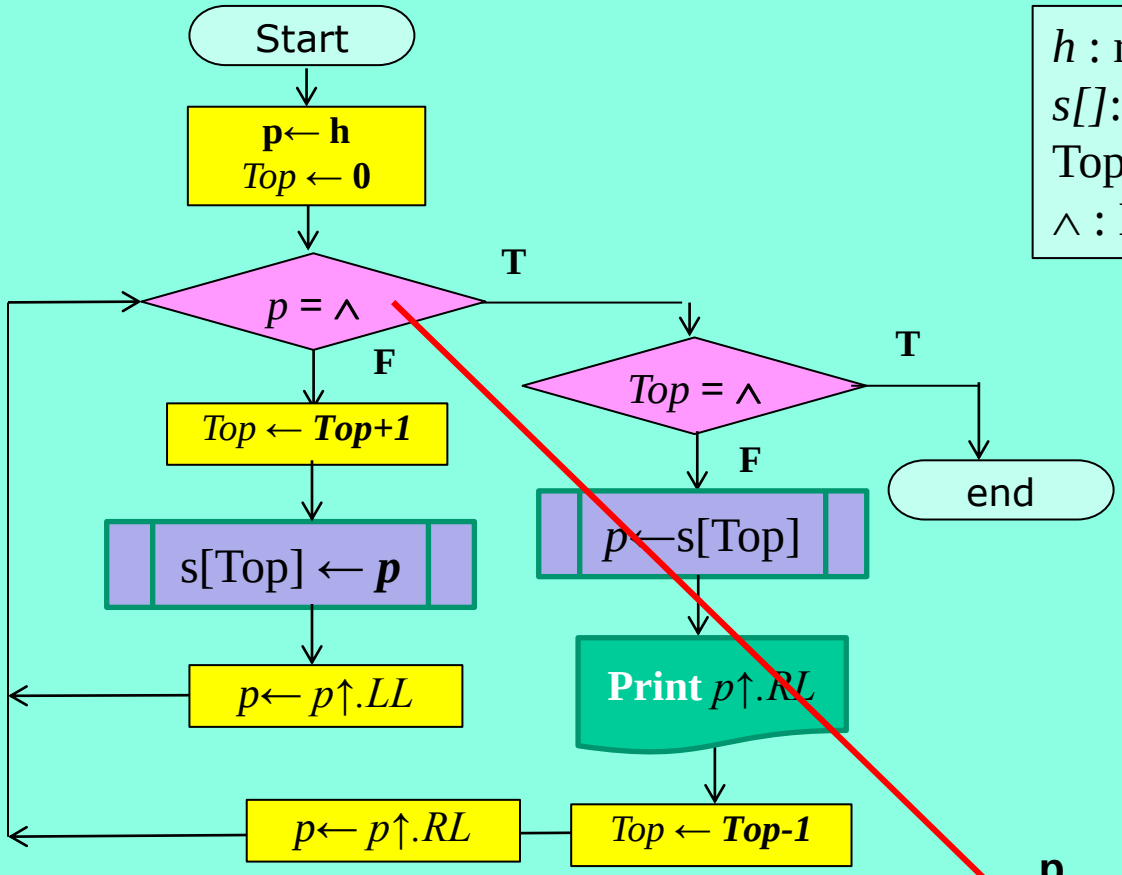

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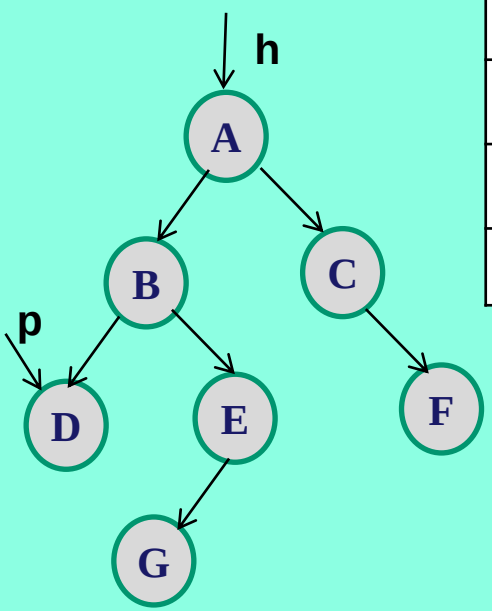

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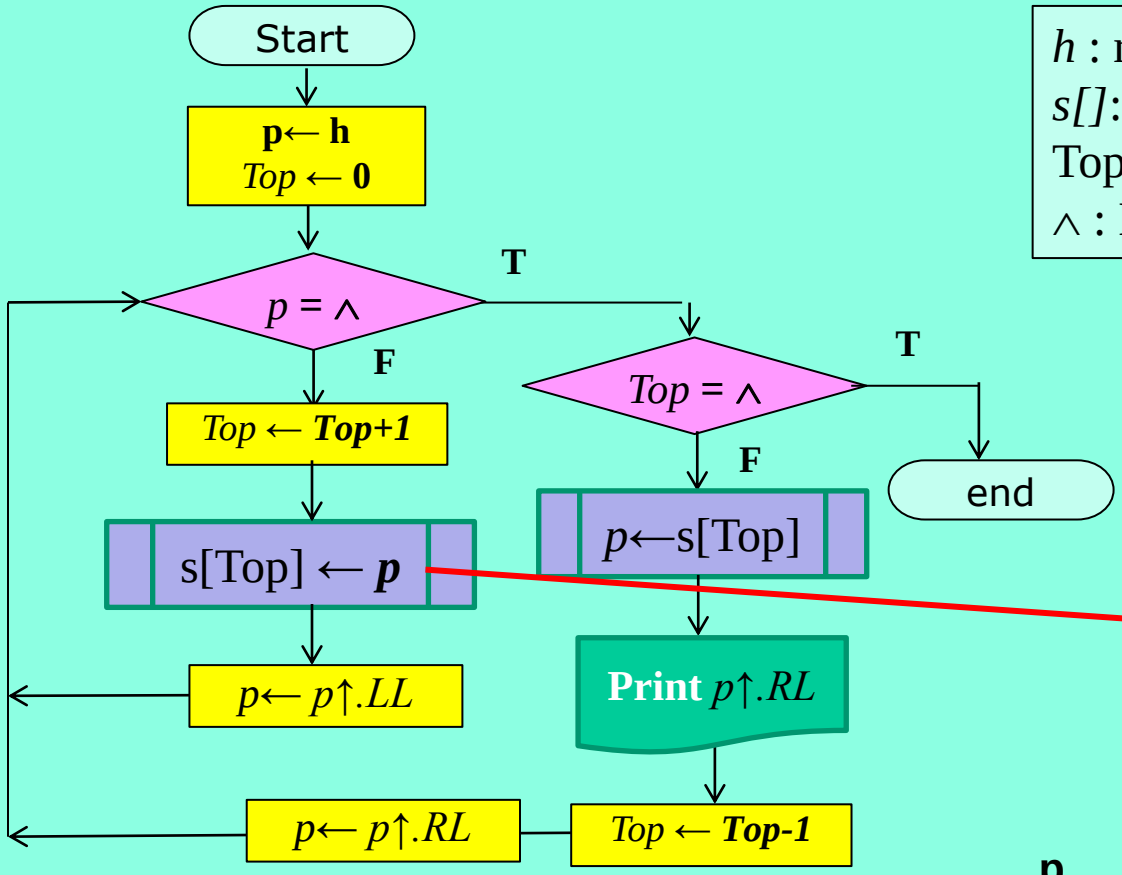

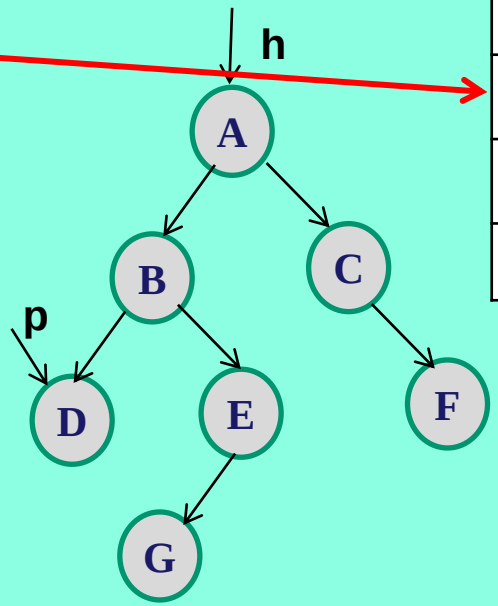

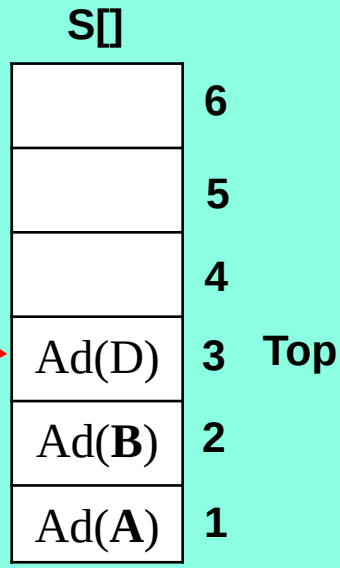

Traversing BT

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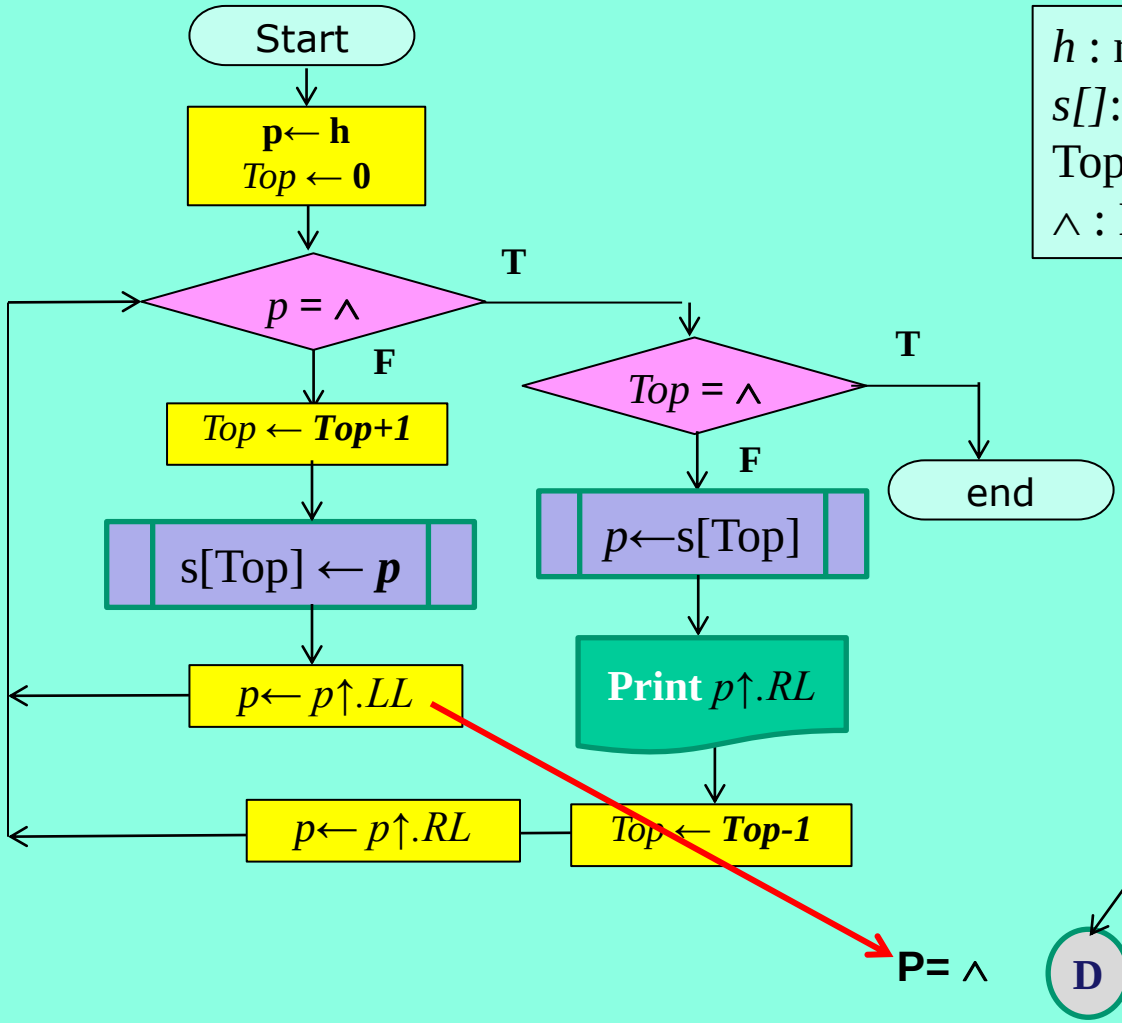

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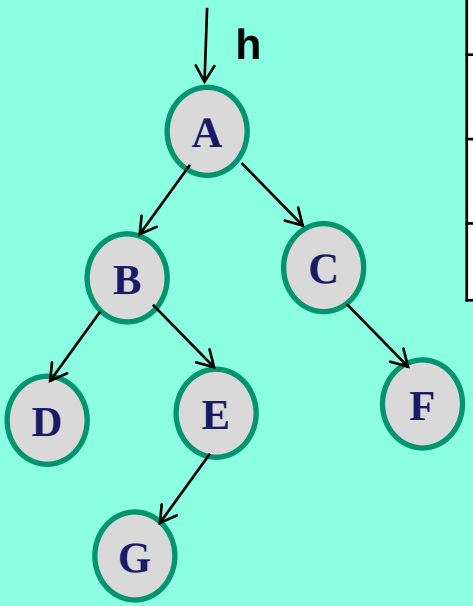
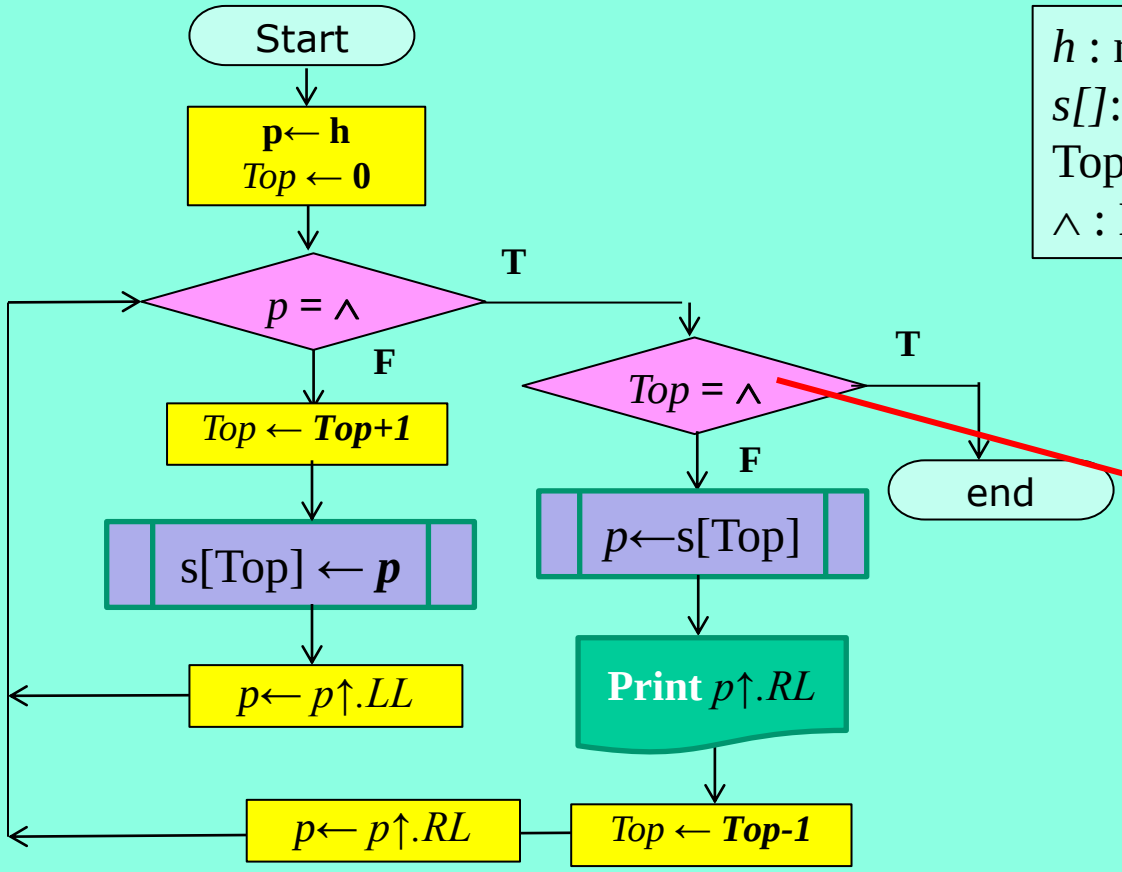


4
1

Traversing BT

Topic 1: Write an Algorithm to traverse a binary tree in inorder traversing.



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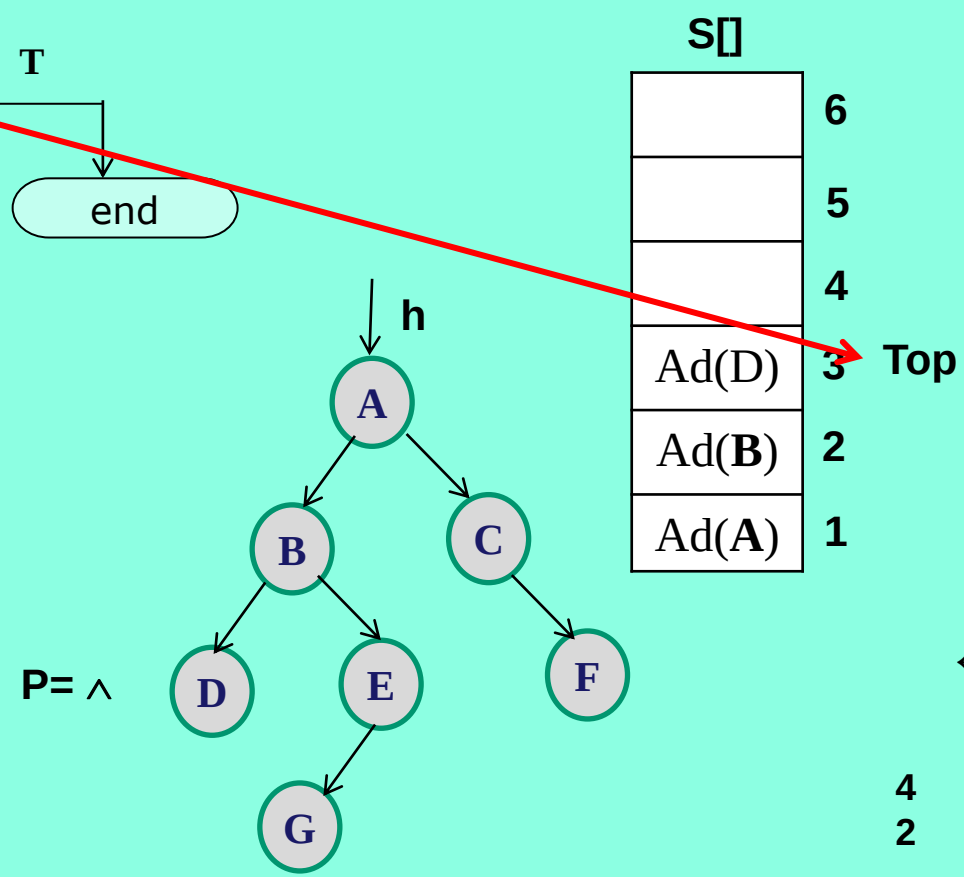

Traversing BT

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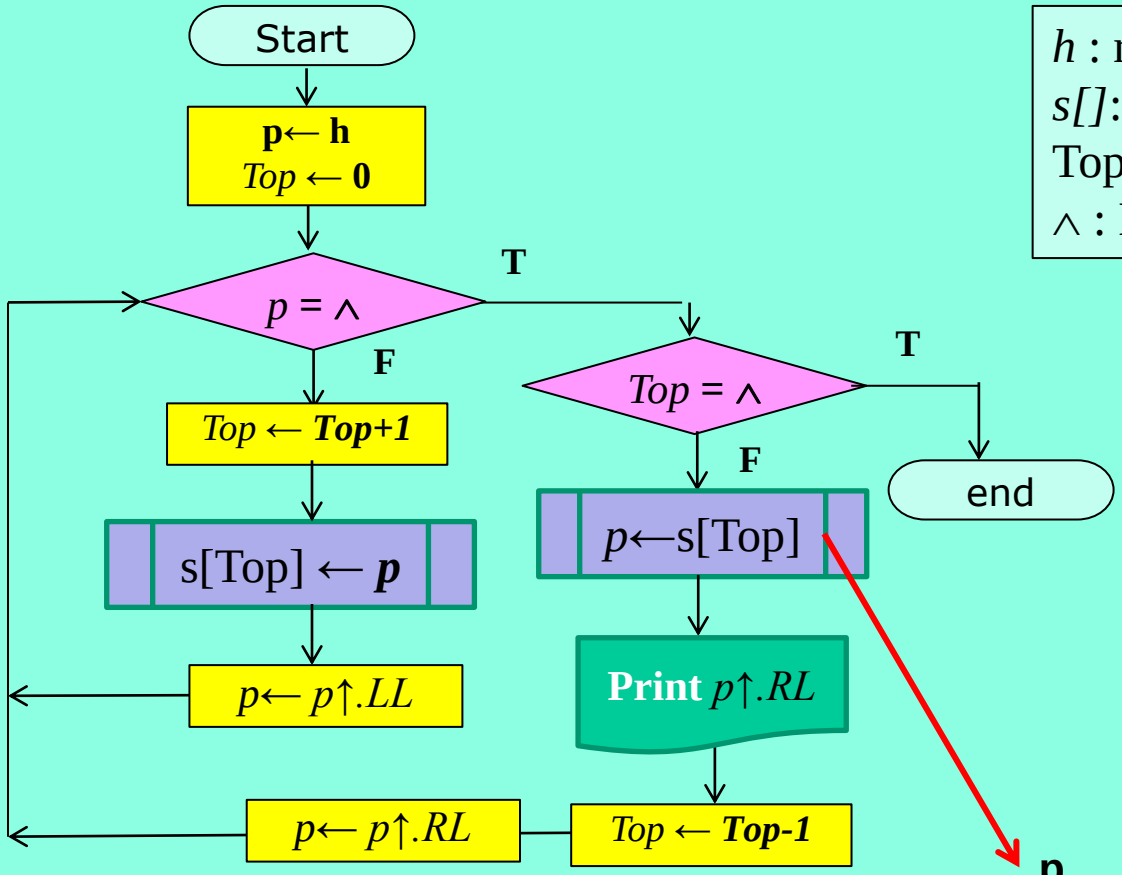

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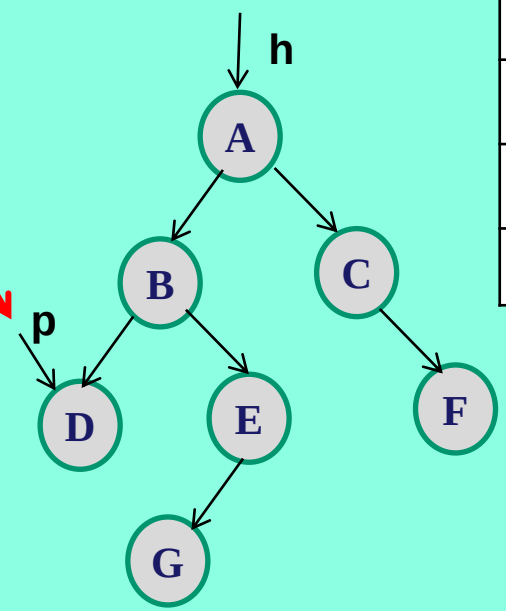

Traversing BT

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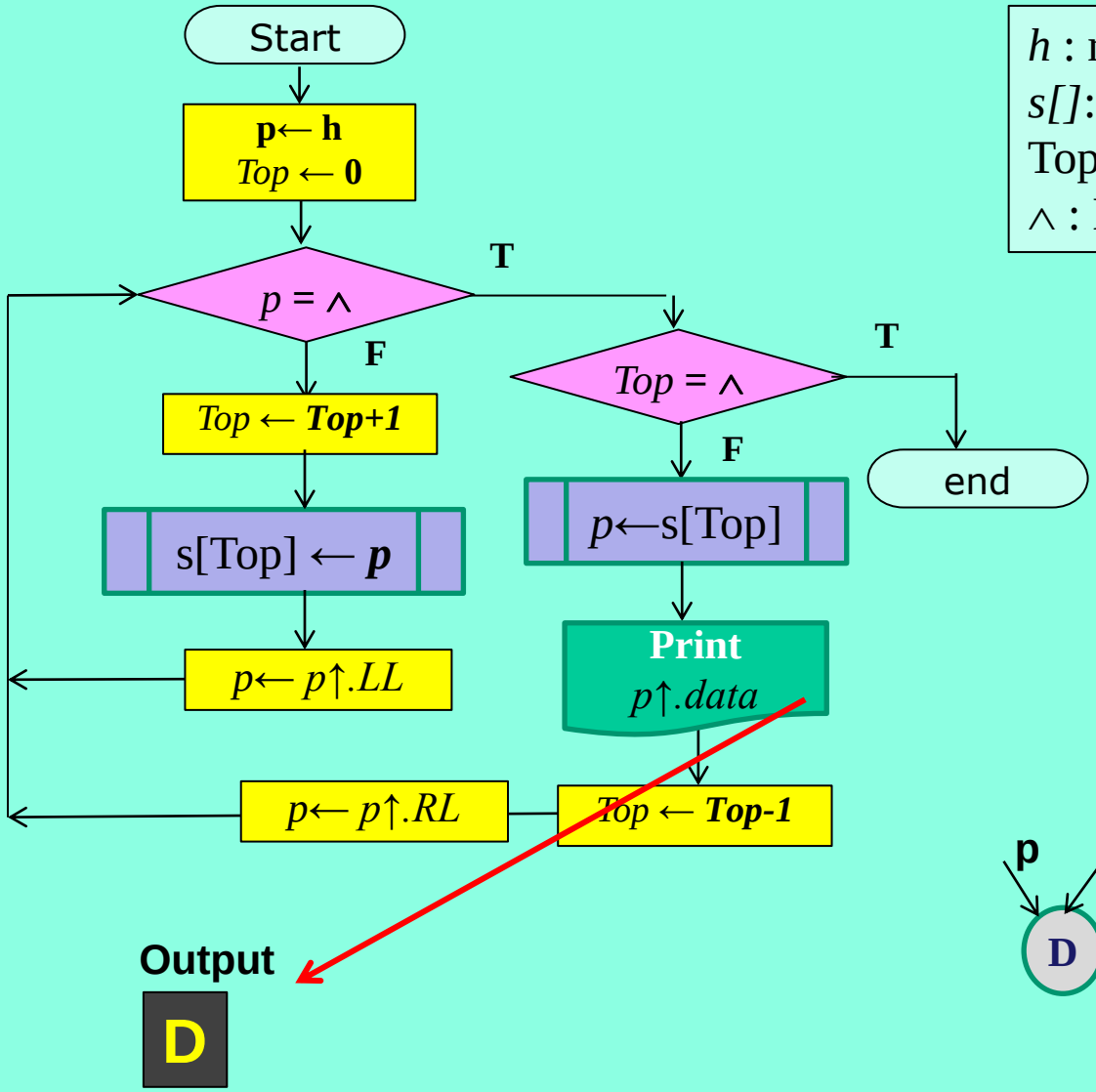

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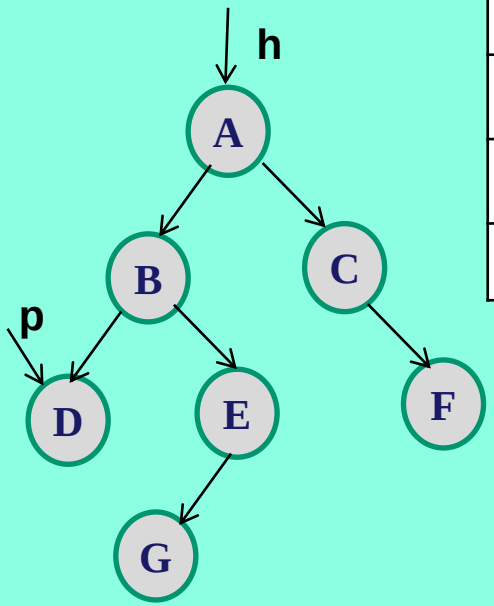
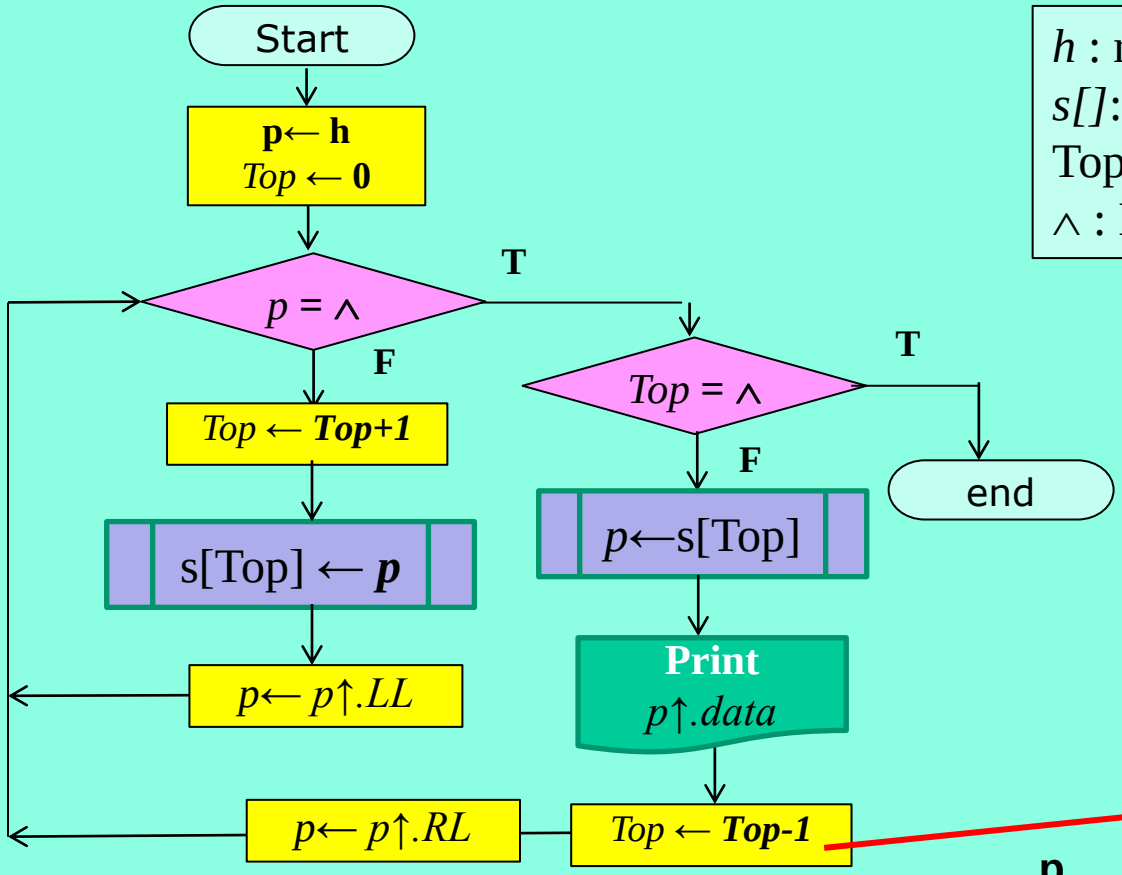


4
4

Traversing BT

Topic 1: Write an Algorithm to traverse a binary tree in inorder traversing.



h : root address
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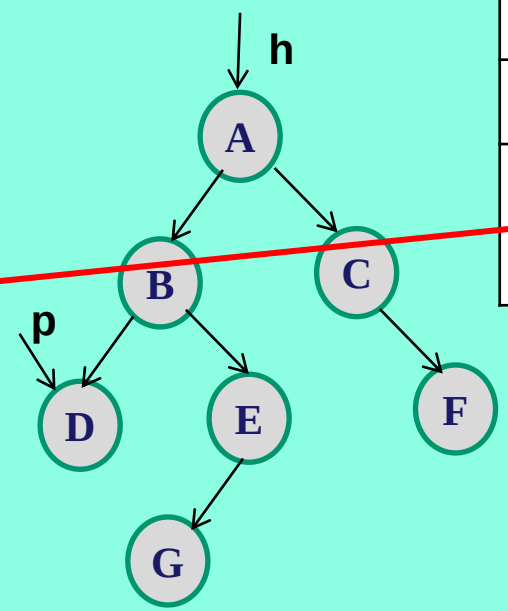
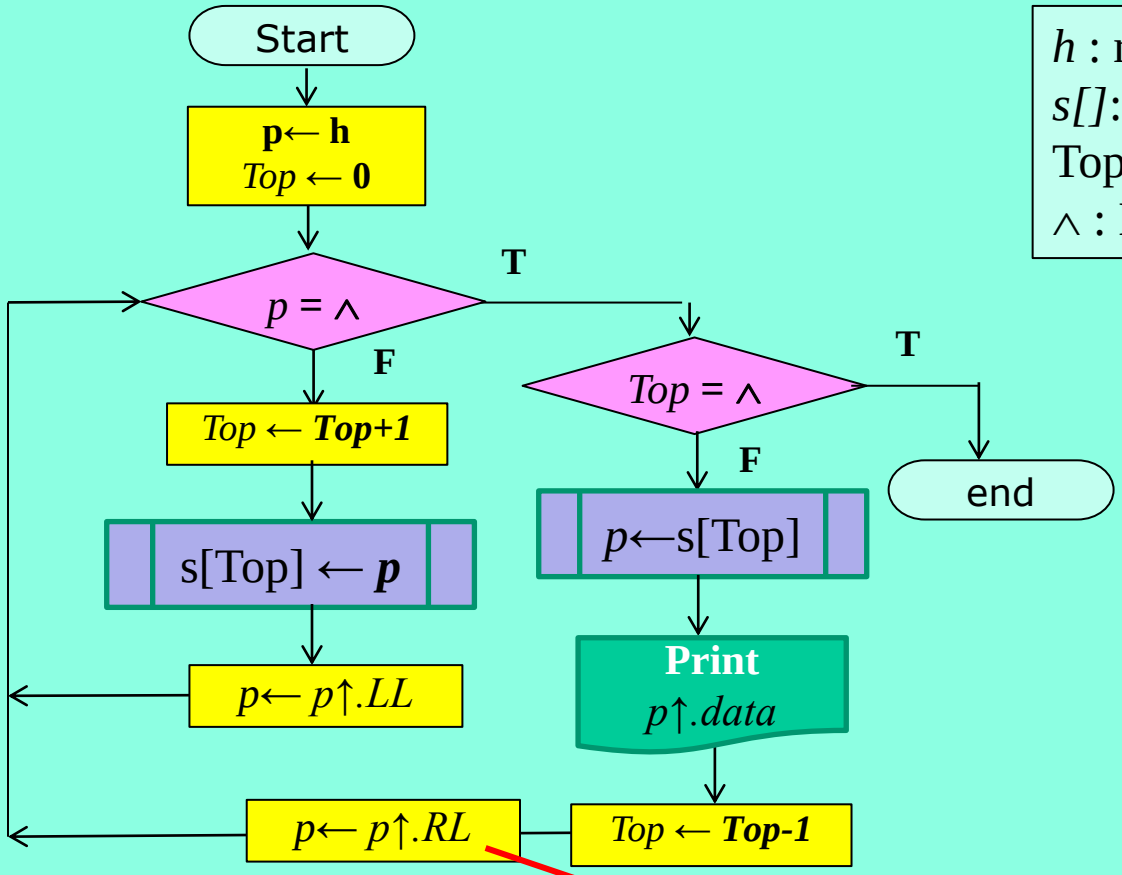


Output

D

Traversing BT

Topic 1: Write an Algorithm to traverse a binary tree in inorder traversing.



h : root address
s[]: stack
Top : Stack Pointer
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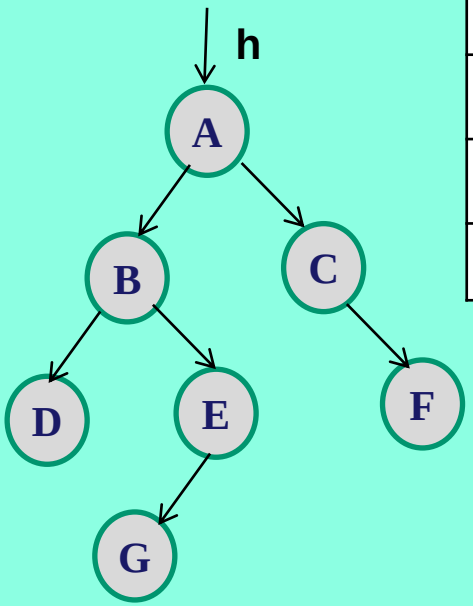

Output

D

4
6

Traversing BT

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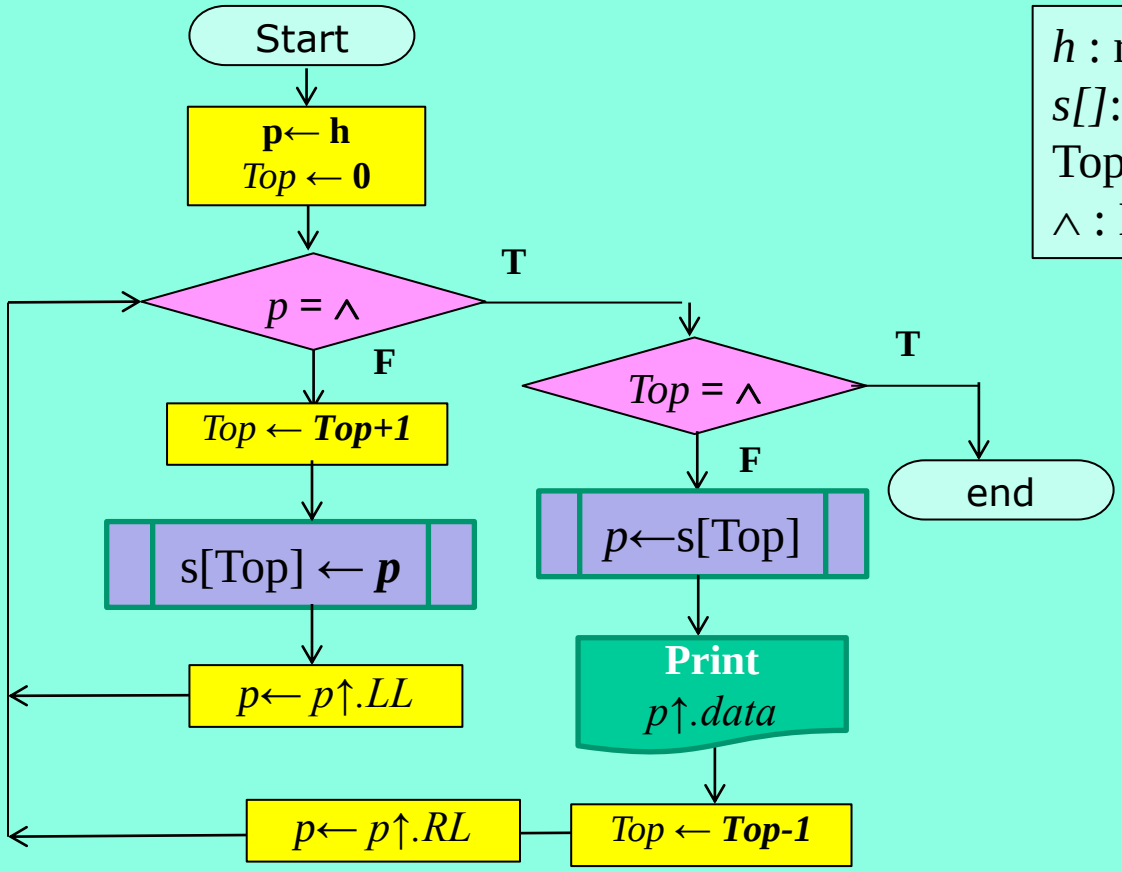

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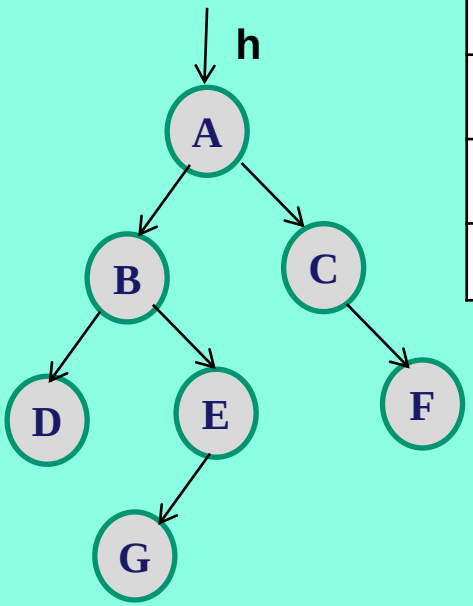
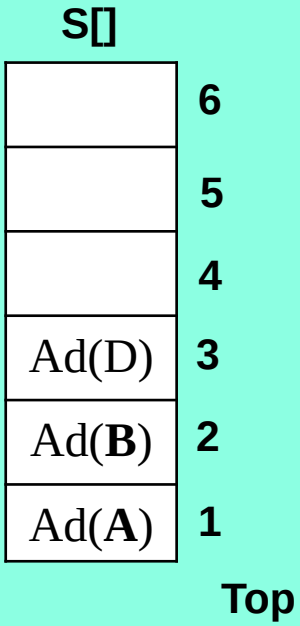


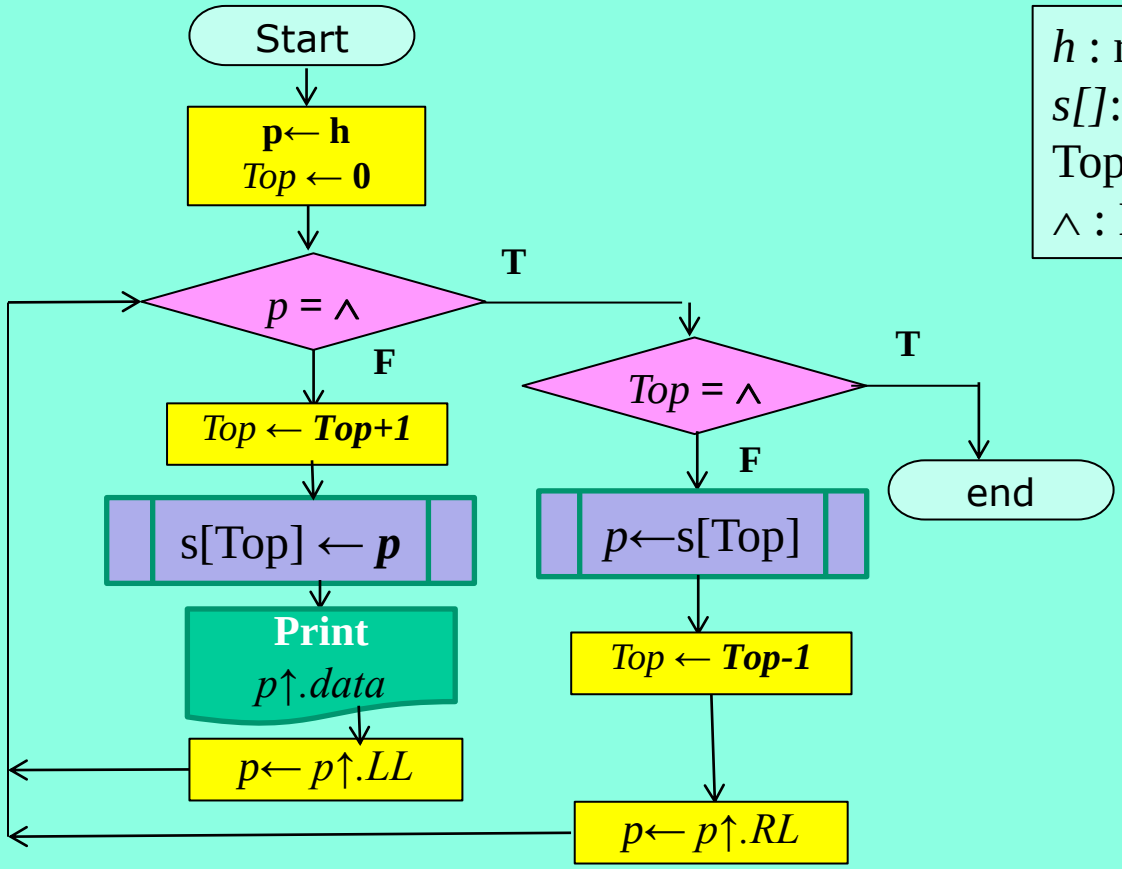


Output

D B G E A C F

Traversing BT

Topic 1: Write an Algorithm to traverse a binary tree in Preorder traversing.



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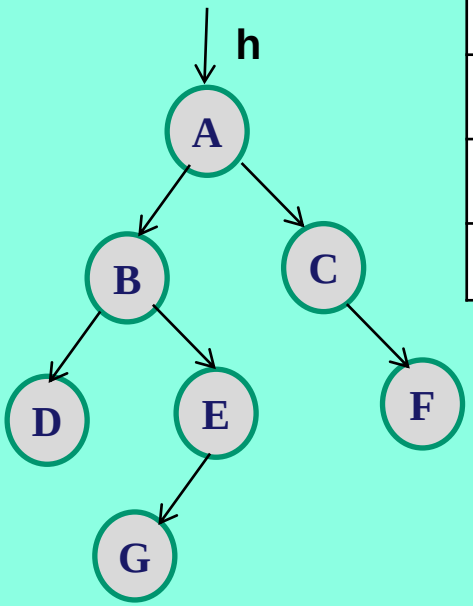

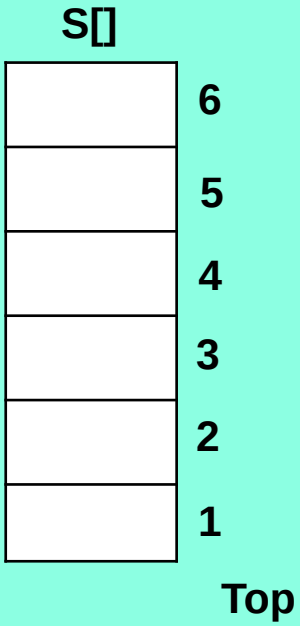

Output

D B G E A C F